

Benchmarking LLM Integration Strategies for MCDA-Assisted Household Energy Decision Support

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Abstract

Residential buildings contribute 38% of U.S. CO₂ emissions, yet most people rely on biased decision making rather than tailored choices. Multi-Attribute Value Theory (MAVT) offers a rigorous method of Multi-Criteria Decision Analysis (MCDA) to reduce emissions, but building MCDAs daily requires expertise that limits household adoption. This study evaluated whether LLMs can bridge that gap by comparing three architectures: $\mathcal{A}_D$ (Pure Prompting), $\mathcal{A}_E$ (RAG-Enhanced), and $\mathcal{A}_H$ (LLM-Parameterized Reference Scoring, which extracts engineering parameters for a deterministic physics calculator). Ground truth calculators were built to benchmark MAVT scores across HVAC, appliance, and shower decisions. $\mathcal{A}_H$ achieved a Kendall’s tau of 0.902, MAE of 0.051, and 91.7% Top-1 accuracy across 195 scenarios. $\mathcal{A}_E$ scored 0.339, 0.138, and 51.7%; $\mathcal{A}_D$ scored 0.095, 0.219, and 35.4%.

Keywords: Large language models, Multi-criteria decision analysis, Multi-Attribute Value Theory, Household energy, Decision support systems, Retrieval-Augmented Generation, Sustainability

1. Introduction

The average U.S. household emits roughly 8,744 pounds of CO₂e and consumes about 10,791 kWh of electricity each year [1, 2]. However, there is more than enough evidence that this could be much further optimized to reduce cost to families and environmental impact [3, 4].

Tools that combine optimization, behavioral insight, and decision support can close part of that gap, but only if they are usable without specialized training. Certain frameworks, such as Multi-Criteria Decision Analysis (MCDA), can be used to choose an optimal balance of factors that matter to users [5]. Unfortunately, these are extremely taxing to create oneself accurately even as an expert, and they require extensive time which makes using them by themselves impractical for household decisions.

However, it has been proposed in the past that LLMs could be used to automate this process, thus making it practical for household decisions. Without knowing how accurate these LLMs are as experts, it is unclear whether this would be a beneficial or detrimental idea.

Modern LLMs carry general knowledge about the energy use and carbon intensity of common products and behaviours, and many systems can retrieve additional information at inference time. They have not, however, been rigorously evaluated for individual-level household decisions against a deterministic, physics-based ground truth, nor has it been established which way of integrating an LLM into the MCDA workflow—direct prompting, retrieval-augmented prompting, or LLM extraction feeding a deterministic calculator—best reproduces that ground truth. This study addresses that gap by benchmarking the three integration strategies head-to-head across three household energy decision types, holding the model and the MAVT weighting constant so that the integration strategy is the variable under test.

This study addresses two research questions:

1. **RQ1:** Which LLM integration architecture— $\mathcal{A}_D$, $\mathcal{A}_E$, or $\mathcal{A}_H$ —most accurately replicates physics-based MAVT scoring for household energy decisions?

2. **RQ2:** How consistently does the architecture ranking hold across models of varying capability?

2. Literature Review

2.1. Household Energy Decision-Making and Adoption Barriers

When transportation, food, and goods are folded in, per-capita footprints rise to about 17.9 tons of CO₂e and 81,767 kWh of total annual energy [6, 7]. The U.S. Department of Energy estimates that households can save \$200–\$400 annually by adopting ENERGY STAR practices [4, 8]. A persistent gap between intent and behavior keeps many households from realizing those savings and footprint reduction[9].

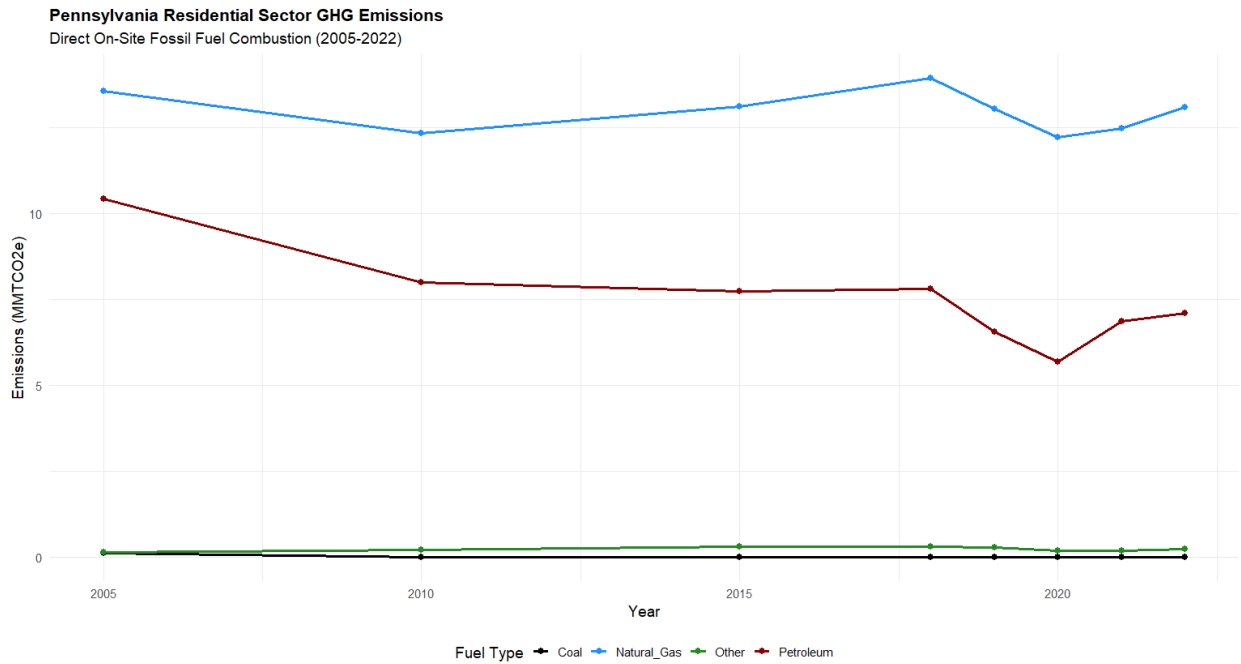


Figure 1: Residential emissions, 2005–2022 (EIA, 2022) [10, 11].

Attari et al. (2010) found that consumers undervalue energy use and savings by a factor of 2.8 on average, systematically underestimating high-energy activities while overestimating low-energy ones [12]. Energy labels partially address this, but effectiveness depends on framing: Newell and Siikamäki (2014) found willingness to pay for efficient products rose when savings were clearly displayed, while Allcott and Taubinsky (2015) reported a 12% market-share increase for compact fluorescent bulbs when benefits were made visible—though gains collapsed once cost projections became ambiguous [13, 14]. Label design also matters: the EU label underperformed relative to U.S. equivalents owing to poor clarity and limited consumer trust [15].

While the average U.S. household consumes 10,791 kWh annually, this consumption is not evenly distributed. Space heating accounts for approximately 43% of residential energy use, followed by appliances (13%), water heating (12%), and air conditioning (8%) [10]. This distribution highlights key intervention points for emissions reduction. National reform of certain key household practices could reduce national emissions by 123 million metric tonnes of carbon (MtC) per year. As of 2005, residential emissions accounted for 38% of overall U.S. CO₂ emissions, totaling 626 MtC. Additionally, these changes would result in little to no reduction in satisfaction, a necessary factor in widespread implementation [3].

Figure 2 reproduces the Dietz et al. display of behavioral plasticity, net carbon effect, and the Reasonably Achievable Emission Reduction (RAER) per action.

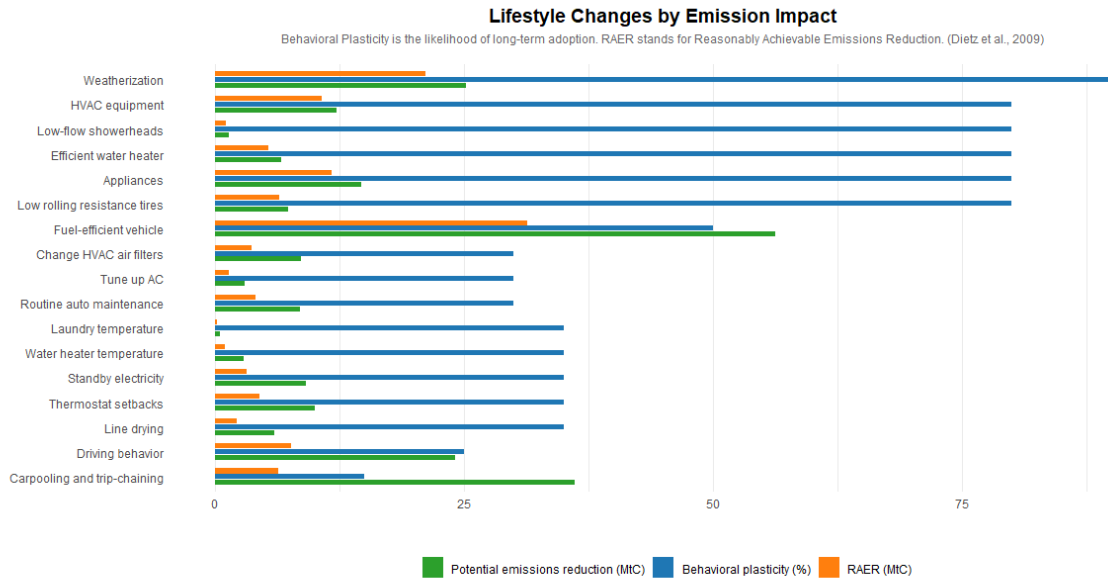


Figure 2: Household behavior changes and their emissions impact (after Dietz et al., 2009 [3]).

Dietz et al. [3] also demonstrated that household-level behavioral changes, if widely adopted, could cut U.S. emissions by 123 MtC annually. However, concerns for behavioral plasticity—how likely people are to implement each action based on financial and social incentives—remain an important factor [3]. Actions such as installing programmable thermostats demonstrate high emissions reductions and adoption likelihood, whereas vehicle upgrades show reductions but low likelihood. More often than not, individuals pointed to curtailment rather than implementing new products for energy efficiency in their households [12]. Thus, targeting simple, heuristic actions is likely to be more effective at this scale than recommending car or HVAC replacements, even if those may be more impactful.

A practical obstacle for MCDA-style tools at the household level is the workflow tax they impose on users. Even when the benefits are clear, switching from a routine choice to a model-mediated decision adds time and cognitive cost, both of which are documented deterrents to adoption [16, 17, 18]. Most household decisions in this space, including thermostat setpoints, appliance run times, and shower length, are heuristic rather than analytical.

Heuristic decision-making is efficient for everyday choices but degrades in complex or high-stakes settings. Under time pressure, people default to satisficing, choosing the first option that clears a minimum threshold, or to dominant prior behaviors, even when better alternatives exist [19]. When time pressure combines with information overload, judgment error rates rise meaningfully [20, 21]. Household energy decisions typically occur in exactly this regime: short deliberation windows, competing demands, and incomplete information. Tools that require sustained attention or structured inputs are unlikely to influence behavior under those conditions.

Attari et al. found that consumers undervalue energy use and savings by a factor of 2.8 on average, with small overestimates for low-energy activities and large underestimates for high-energy activities [12]. Allcott and Taubinsky found that willingness to pay increased by \$2.30 for compact fluorescent lights when benefits were displayed, and market share rose 12%. They proposed this would extend to most household appliances. Once cost-savings statistics became unclear, the impact of all information dropped significantly [14]. Any tool recommending decisions should account for energy cost and display projected changes to motivate users to follow through.

However, a nuance to that is the compounding obstacle of temporal discounting: the perceived value of a future reward decreases with delay in a non-uniform way. People prefer \$100 today over \$110 tomorrow, yet prefer \$110 in a year to \$100 in eleven months, suggesting the discount operates sharply over short intervals [22]. The benefits of reducing energy consumption are largely abstract and deferred; even tools that promise genuine future savings may

be ignored if those benefits are not immediate, concrete, or personally meaningful.

Multi-criteria decision analysis (MCDA), life-cycle assessment (LCA), and Pareto-based prediction methods have been used to track and reduce environmental impact at building and organizational scales [5]. Engineering frameworks of this kind can optimize complex decisions, but their construction is time-consuming: signal-processing pipelines such as the Hilbert–Huang transform are not rapid, and constructing the decision matrix itself is a recognized bottleneck [23, 5]. Huang and Burgherr (2024) addressed part of this by building a streamlined MCDA calculator that compresses several accepted methods into a single workflow [24]. Even with their simplification, the user still needs enough familiarity with MCDA construction to frame the alternatives, criteria, and weights, which keeps the tool out of routine household use.

The harder problem is operational. Constructing an MCDA framework for a specific decision normally requires methodological knowledge, including the choice of weighting scheme, value function form, and normalization, alongside rubric design and domain familiarity [23, 24]. Existing simplified calculators reduce but do not eliminate this barrier. A large language model that can convert a natural-language description of a household situation into the alternatives, criteria, and contextual parameters that an MCDA needs would, in principle, remove the methodological barrier altogether.

AI is increasingly being used to build or operate decision frameworks in domains that share the household setting’s combination of variability and judgment. Generative AI now appears in operations and supply-chain decision pipelines [25], in marketing strategy [26], and in clinical decision support [27]. These techniques have been used across many different industries, and AI’s ability to manage large datasets suggests it could scale down to personal decision-making as well.

Specific AI techniques integrated into MCDA pipelines include fuzzy systems [28], which account for uncertainty and degrees of truth to approximate human reasoning. Fuzzy anchors could guide a model given no outside knowledge, such as $\mathcal{A}_D$.

The plausibility of this approach is reinforced by recent work outside the energy domain. Svoboda and Lande (2024) used GPT-4 to construct an analytic hierarchy process (AHP) decision model for cybersecurity, reporting a consistency ratio below the conventional 0.10 acceptability threshold and noting that earlier models such as GPT-3 were unreliable [29]. That result indicates that LLMs above a certain capability level can produce coherent MCDA structures, but it leaves open the separate question of whether they can produce accurate quantitative scores against a physics-based ground truth.

2.2. LLMs as Numeric Reasoners and Scorers

LLMs are autoregressive next-token predictors, not symbolic computation engines. Cobbe et al. (2021) introduced the GSM8K benchmark of grade-school arithmetic word problems and found that even large Transformers fall well short of human accuracy once a problem needs more than a few chained steps, with errors compounding rather than cancelling [30]. The gap hasn’t closed in newer models: Li et al. (2025) test GPT-4, DeepSeek, and others on basic arithmetic, magnitude comparison, and multi-step numeric reasoning, and report the same weaknesses despite strong general-language performance [31]. Imani et al. (2023) approach the same problem from the fix side, building MathPrompter on the observation that LLMs “often provide incorrect answers” on arithmetic benchmarks unless cross-checked against multiple generated solution paths [32]. Approximating a tent-shaped comfort function, applying a multiplicative budget penalty, and tracking a TOU emissions schedule, then reporting a single calibrated number, is exactly this kind of task – so unaided LLM scoring (the $\mathcal{A}_D$ architecture here) carries a structural rather than incidental risk of numeric drift. The canonical use of Retrieval-Augmented Generation is injecting external facts into an LLM’s context to reduce hallucination in open-domain QA [33]. An MCDA scoring task is closer to Case-Based Reasoning, which treats problem-solving as retrieving similar prior cases and reusing their recorded solutions [34]. Wiratunga et al. (2024) build CBR-RAG, where retrieved legal cases act as structured exemplars that shape an LLM’s reasoning rather than supply facts [35]; Minor and Kaucher (2024) get a similar result, finding that retrieved example explanations used as in-context demonstrations improve LLM-generated outputs over zero-shot prompting [36]. Read against this, the $\mathcal{A}_E$ architecture’s exemplars look less like supplementary facts and more like a case-based rubric: scenarios whose ground-truth scores anchor the model’s numeric judgment to the decision space’s actual reference distribution, rather than whatever scale it absorbed during pretraining. Scoring instead of solving doesn’t dodge the problem, either. Wang et al. (2024) show that LLMs exhibit positional bias when scoring, ranking the first response

Symbol	Definition
τ	Kendall’s rank correlation coefficient
ρ	Spearman’s rank correlation coefficient
w_i	weight of criterion i ($i = 1 \dots 4$), summing to 1
$v_i(x)$	value function score for criterion i at raw value x , range $[0, 10]$
α	shape parameter for logarithmic value functions
s_j	MAVT weighted sum score for alternative j
MAE	Mean Absolute Error on MAVT score (0–10 scale)
RMSE	Root Mean Squared Error on MAVT score (0–10 scale)
n	number of test scenarios evaluated
k	number of retrieved chunks per query in RAG ($k = 3$)

Table 1: Notation used throughout this paper.

higher simply because of its placement [37]. Their conclusion that LLMs are “not fair evaluators” in default zero-shot configurations [38] motivates our study rather than contradicts it: we test whether more capable models, deployed within a structured MCDA tool rather than as unaided judges, can overcome the weaknesses prior work identified. Li et al. (2026) document systematic scoring-bias distortions in LLM-as-a-judge pipelines that persist even in advanced models [39], and Zhu et al. (2026) trace most of the cross-run score variance on benchmarks like MT-Bench to judge identity rather than response quality [40]. The pattern isn’t confined to judge benchmarks: Johnson and Zhang (2024) find GPT-4o essay scores clustering well below human ratings with noticeably less variance—central-tendency bias, where the model avoids extreme scores [41]—and Poličar et al. (2025) find that different LLM graders carry distinct, model-specific leniency or strictness profiles relative to human TAs in a bioinformatics course [42]. For $\mathcal{A}_D$ specifically, this predicts a clean failure mode: cluster the four criterion scores for all three alternatives into a narrow middle band, and the sharp, non-linear penalties in the ground-truth value functions—the shower calculator’s tank-capacity cutoff, the budget penalty’s exponential tier—get missed entirely, dragging down Kendall’s tau even when the scores are directionally reasonable.

Despite extensive work on LLM numeracy, retrieval-augmented generation, and LLM-as-judge scoring, no prior study has rigorously compared these integration strategies against a deterministic, physics-based ground truth for household energy decisions. The present study fills this gap by benchmarking three LLM-MCDA architectures— $\mathcal{A}_D$, $\mathcal{A}_E$, and $\mathcal{A}_H$ —across 195 test scenarios spanning HVAC, appliance, and shower decisions, using identical MAVT weighting and model access so that the integration strategy is the sole variable under test.

Section 2 reviews the relevant literatures on household energy behavior and LLM numeric reasoning. Section 3 describes the MAVT framework, ground-truth calculators, three architecture implementations, and evaluation metrics. Section 4 presents results across ranking accuracy, criterion-level error, decision types, and a RAG ablation. Section 5 discusses limitations and future work; Section 6 concludes.

3. Methodology

3.1. Decision Types and Scenario Corpus

Three decision types were selected for their high behavioral plasticity and RAER: HVAC (thermostat setpoints), Appliance (time-of-use scheduling), and Shower (duration).

The scenario corpus has two disjoint pools. The *test set* contains 195 scenarios (70 HVAC, 65 Appliance, and 60 Shower); each scenario presents three alternatives scored independently by each architecture and by the ground-truth calculator. The *RAG corpus* contains an additional 90 scenarios (35 HVAC, 35 Appliance, and 20 Shower) that populate the retrieval index used by the $\mathcal{A}_E$ architecture; RAG scenarios are never used for evaluation [33].

The 195 test / 90 RAG split provides a 2.17:1 test-to-RAG ratio sufficient for retrieval coverage; HVAC is overrepresented (70 test + 35 RAG) due to its higher parameter dimensionality (R-value, SEER, age, occupancy, insulation tier).

The tables below detail the parameters given to architectures vs. calculators per decision type.

Table 2: HVAC parameters: Architectures vs. calculator

Parameter	LLMs	Calc.
Location	✓	—
Outdoor Temp	✓	✓
Home Size (sq ft)	✓	✓
Insulation	✓	—
R-Value	—	✓
Household Size	✓	✓
Housing Type	✓	✓
House Age ^a	✓	✓
Utility Budget	✓	✓
SEER Rating	—	✓
Occupancy Context	—	✓
Indoor Temp (from alternative)	—	✓

^aLLMs receive a building-age range label (e.g., “20+ Years”); the calculator uses HVAC equipment age in exact years — distinct concepts.

Table 3: Appliance parameters: Architectures vs. calculator. A ✓ in the LLMs column denotes a field explicitly present in the prompt; appliance type and baseline time are conveyed implicitly through the natural-language question text (“run the dryer,” “it’s around 7 in the evening”) rather than as labeled fields. TOU period, utility rate, kWh/cycle, cycle duration, and monthly-cycles are calculator-internal and are not shown to the architectures.

Parameter	LLMs	Calc.
Appliance Type	(text)	✓
Baseline / Run Time	(text)	✓
Appliance Age Range	✓	—
kWh per Cycle	—	✓
TOU Period / Utility Rate	—	✓
Monthly Cycles Assumption	—	✓
Household Size	✓	✓
Housing Type	✓	✓
Location	✓	✓
Utility Budget	✓	✓

The calculator receives specific engineering values that would require professional measurement for households, and thus are unreasonable to be requested of homeowners. The architectures receive generalized version (e.g. insulation, which is a general estimate of poor/average/good rather than a specific R-value) or do not receive counterparts to some parameters at all (e.g. occupancy context, which the architecture instead has to infer from the question text). These parameters are chosen as they are things a household member would actually know [43].

To generalize parameters, we identified acceptable ranges for each qualitative level from published standards and mapped each scenario to a representative value within that range. Insulation tiers map to standard U.S. residential wall R-values: Poor to R-11 (minimal 2×4 cavity fill, below current code), Medium to R-13 (standard 2×4 batt), and Good to R-19 (2×6 framing), consistent with code tables for wood-frame assemblies and ENERGY STAR guidance [44, 45]. For the Flow Rate parameter, the scenario’s numeric GPM is bucketed into three qualitative labels: `low_flow` (≤ 2.0 GPM, including EPA WaterSense certified 1.5 GPM showerheads [46]), `standard` (2.5–3.0 GPM, at or near the Energy Policy Act of 1992 federal maximum of 2.5 GPM [46]), and `high_flow` (> 3.0 GPM). The label is shown to the LLM; the calculator always receives the exact numeric GPM from the scenario. Table 5 summarizes the generalized parameters across all three decision types. The two sets overlap on environmental conditions, however so that both pipelines are reasoning about the same physical scenario. The LLM never sees the engineering parameters directly. For the $\mathcal{A}_H$ architecture, which needs to use those parameters within its calculations, it infers structured estimates of them from the description before the calculator is invoked. Further detail on choosing these specific parameters and their effects is available in section 2.2.

Table 4: Shower parameters: Architectures vs. calculator.

Parameter	LLMs	Calc.
Location	✓	—
Outdoor Temp	✓	✓
Inlet Water Temp	—	✓
Heater Setpoint	—	✓
Flow Rate (low_flow/standard/high_flow)	✓	—
GPM	—	✓
Duration (min)	(alt.)	✓
Household Size	✓	✓
Tank Size	—	✓
Utility Budget	✓	✓
Housing Type	✓	✓

Duration is the alternative under evaluation (one of the three the architecture scores), not a separate context field; heater setpoint, tank size, GPM, and inlet temperature are calculator-internal and withheld from the architectures.

Table 5: Parameters generalized from engineering values to homeowner-accessible labels. LLMs receive the label; calculators receive the numeric value. R-values in $\text{h}\cdot\text{ft}^2 \cdot ^\circ\text{F}/\text{BTU}$; GPM dataset range 1.5–4.0; kWh/cycle ranges span the full dataset.

Parameter	LLM Label	Calc. Value	Source
<i>HVAC</i>			
Insulation	Poor	R-11	[44, 45]
	Medium	R-13	
	Good	R-19	
<i>Shower</i>			
Flow Rate	low_flow (≤ 2.0 GPM)	scenario GPM	[46]
	standard (2.5–3.0 GPM)	scenario GPM	[46]
	high_flow (> 3.0 GPM)	scenario GPM	
<i>Appliance</i>			
Appliance Age	1–15 yr	Dishwasher: 0.72–1.70 kWh/cyc Washer: 0.15–0.45 kWh/cyc Dryer: 1.15–3.50 kWh/cyc	[47, 48, 49]

3.2. MAVT Framework Design

The Multi-Attribute Value Theorem (MAVT) allows alternatives to be compared on different criterion based on the range of values that occurs for each criterion. In our study, we chose the following weights: Environmental Impact (35%), Energy Cost (30%), Comfort (20%), and Practicality (15%). The weights are held constant across all decision types, all architectures, and the ground-truth calculator. We did this so that any difference in score is because of the inputs and value function, not because of difference in the way each criterion is perceived for the three decision types. Normally, MAVT additive form (Eq. 1) requires preferential independence among criteria [50, 51]. This means that criteria should be independent, meaning that the value (good or bad) of one criterion should have no effect on the value of another criterion.

$$s_j = \sum_{i=1}^4 w_i \cdot v_i(x_{ij}) \quad (1)$$

In the case of our criteria, the energy cost and the environmental impact criterion are not completely independent, as they both stem from the kWh usage of a certain activity. The Spearman’s rho between energy cost and environmental impact across the full scenario set is 0.92, which does indeed mean that they are correlated. However, this is a feature of the dataset and of household emissions, and it is actually possible for the two criterion to diverge. Since environmental impact comes from actual CO₂, which varies based on the type of power plant generating the power, grid mix composition, renewable penetration, and regional interconnection flows [52], and energy cost comes from the time of day, gas peakers, demand charges, transmission congestion pricing, and utility revenue requirements [53]. While Pennsylvania does not usually have such divergence, there are other areas that do, such as Texas (ERCOT), where high daytime wind generation can drive electricity prices near zero while gas-fired backup units sustain elevated marginal emissions during evening demand peaks [54, 55].

The Environmental weight of 0.35 reflects two convergent considerations. Residential emissions account for a sizable share of national CO₂ emissions [3], so a household decision-support tool that does not give priority to environmental impact fails. The Value-belief-norm theory, which says that show that environmental orientation is the most consistent normative driver across pro-environmental household behaviors [56, 57, 58, 59, 60, 61, 62, 63]. Finally, anyone using such a tool obviously has environmental motivations, and entropy-based normalization of the scenario distributions assigns higher information weight to the environmental criterion, independently supporting an above-average allocation [64].

Cost is the strongest independent predictor of consumer adoption. This is clear for multiple reasons. First, cost savings as shown on energy labels for consumer products were more effective than actual energy use and carbon emissions.[13]. When energy labels strip away concrete cost-savings figures, the influence of all remaining information drops sharply [14, 15]. Second, households misestimate energy use, with Attari et al. reporting that participants underestimated energy use and savings by a factor of 2.8 on average [12]. Thus, high importance must be given to savings, even if the user is automating the process. It would also be beneficial to directly display actual cost savings. Weighting Energy Cost at 0.30 keeps it slightly below environmental impact while keeping its role as the criterion households actually act on. For HVAC decisions, thermostat setpoints directly determine electricity draw at the billing rate; households shown real-time cost feedback respond by reducing peak-period consumption, confirming that cost remains the operative signal even for automated decisions [65]. For shower decisions, water heating accounts for a significant share of residential energy costs and the per-shower increment is rarely visible to users, amplifying the same underestimation bias and making cost-feedback interventions similarly effective [66, 67].

Comfort is weighted at 0.20. Thermal comfort is the dominant driver of HVAC behavior even when it conflicts with energy savings [68, 69, 70], and low-income households in particular trade comfort against cost in documented ways [71]. Anchoring comfort to ASHRAE Standard 55 [72, 73] keeps the criterion physically interpretable for HVAC and provides a 0–10 scale that generalizes (with domain-specific calibration described in Section 3.3) to appliance delay and shower duration. For appliance decisions, comfort is primarily a function of run-time delay: demand-response research finds that household willingness to accept a shift decays sharply beyond two to four hours, with dishwashers tolerated longest and dryers shortest [74, 75], and the penalty scales further with household size as dishes and laundry accumulate faster. For shower decisions, comfort peaks around the REU2016 average of 7.8 minutes [66, 67] and depends additionally on temperature adequacy, with heater setpoints below CDC Legionella thresholds carrying a comfort penalty [76].

Practicality is weighted at 0.15. A persistent finding in the heuristics literature is that any tool that adds friction relative to the existing routine sees adoption drop sharply, and heuristic decision-making remains the default for everyday energy choices [16, 17, 77, 78]. Penalizing alternatives that violate budget, system, or behavioral limits keeps the tool’s recommendations in check to make sure that households would actually adopt it long term. The lower weight (relative to comfort) reflects that practicality is a constraint on what is reasonable rather than a primary preference dimension. Where comfort captures what a household would prefer given no constraints, practicality captures what is feasible and sustainable to adopt. A 15-minute shower may score near-optimally on comfort, but if four occupants share a 40-gallon tank, available hot water runs out before the third person showers — making that alternative infeasible regardless of comfort preference.

By requiring a score or parameter estimate for each of the four criteria, the structured framework prevents the LLM from arbitrarily selecting an alternative based on a single dominant attribute, a known failure mode of unaided LLM judgment [38].

A sensitivity analysis was conducted to test the stability of the results. Fluctuations of ± 0.05 were applied to the weight vector by perturbing each of the four criterion weights individually and redistributing the difference equally across the remaining three, plus one equal-weight scenario ($w_j = 0.25$ for all j). The results can be seen in the below table, Table 6.

Table 6: Ten weight perturbation scenarios for the sensitivity analysis. Ene = Energy Cost, Env = Environmental, Com = Comfort, Pra = Practicality. Weights rounded to four decimal places. The architecture ordering ($\mathcal{A}_H > \text{RAG} > \text{Pure}$) is preserved across all perturbations (Section 4.1).

Scenario	w_{Ene}	w_{Env}	w_{Com}	w_{Pra}
Baseline	0.3000	0.3500	0.2000	0.1500
Ene +0.05	0.3500	0.3333	0.1833	0.1333
Ene −0.05	0.2500	0.3667	0.2167	0.1667
Env +0.05	0.2833	0.4000	0.1833	0.1333
Env −0.05	0.3167	0.3000	0.2167	0.1667
Com +0.05	0.2833	0.3333	0.2500	0.1333
Com −0.05	0.3167	0.3667	0.1500	0.1667
Pra +0.05	0.2833	0.3333	0.1833	0.2000
Pra −0.05	0.3167	0.3667	0.2167	0.1000
Equal	0.2500	0.2500	0.2500	0.2500

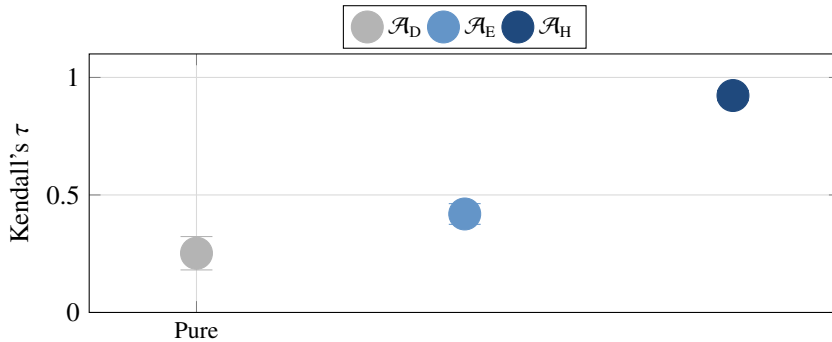


Figure 3: Kendall’s τ range for each architecture across the nine weight-perturbation scenarios (Table 6). The bands are non-overlapping, confirming that the architecture ordering ($\mathcal{A}_H > \text{RAG} > \text{Pure}$) is preserved under ± 0.05 weight perturbations.

The sensitivity analysis (Table 6) confirms that the architecture ordering is preserved under ± 0.05 perturbations of each weight, indicating that the baseline choices are not threshold effects. Figure 3 visualizes the τ range across all nine perturbation scenarios for each architecture: $\mathcal{A}_H$ remains above 0.9, $\mathcal{A}_E$ above 0.37, and $\mathcal{A}_D$ below 0.33, with no

overlap between the architecture bands.

Table 7: Reference ranges (5th–95th percentiles). Shower Environmental Impact is reported in gallons of water; HVAC and Appliance in lbs CO₂.

Criterion	HVAC	Appliance	Shower
Energy Cost (\$)	0.38–3.29	0.025–0.71	0.14–1.14
Environmental Impact	1.96–18.04	0.288–3.643	6.0–45.0
Comfort	0.0–10.0	0.0–10.0	0.0–10.0
Practicality	0.5–10.0	0.5–10.0	0.5–10.0

The reference ranges in Table 7 are computed as the 5th–95th percentiles of the realized ground-truth quantity distributions over the scenario corpus (cost and emissions for HVAC and Appliance; cost and water volume for Shower), rather than extreme theoretical limits, so that normalization preserves meaningful spread across typical residential alternatives. Using dataset percentiles rather than the full theoretical physics envelope is a deliberate modeling choice [64]: it concentrates score sensitivity in the range households actually encounter, at the cost of truncating scores for extreme outlier scenarios (documented as a paper limitation in the Discussion). The endpoints below are the percentile values to two decimals; the corresponding literature anchors are given for interpretability.

HVAC. For HVAC, the cost percentiles are computed over the active-conditioning alternatives only: a setpoint at or near the outdoor temperature (and the bare “Off” option) produces zero decision-period load and collapses to \$0, which would make the lower bound degenerate. Excluding those zero-load points, the realized 8-hour cost distribution at Pennsylvania’s residential rate (\$0.19/kWh) [10, 79] spans \$0.38–\$3.29 (5th–95th percentile); an Off or zero-load alternative still receives the top of the cost score via the decreasing-value normalization. These endpoints remain consistent with residential HVAC efficiency studies (efficient: [80]; degraded: [81]). The HVAC environmental bounds (1.96–18.04 lbs CO₂) are obtained by mapping the same cost-derived kWh envelope through PJM marginal emissions factors (0.976 off-peak, 1.041 peak lbs CO₂/kWh) [52], maintaining a physically consistent link between cost and emissions.

Appliance. For appliances, the realized per-cycle cost distribution (kWh/cycle multiplied by the resolved time-of-use rate across the six PA utilities) spans \$0.025–\$0.71 at the 5th–95th percentile, and the realized per-cycle emissions distribution (kWh/cycle through the PJM marginal factor at run time) spans 0.288–3.643 lbs CO₂. The underlying per-cycle energy endpoints remain consistent with the ENERGY STAR certified-products distribution (efficient HE washers near 0.25 kWh) through older non-certified electric resistance dryers [82, 83, 84, 85, 86, 87].

Shower. For showers, the realized per-shower cost distribution spans \$0.14–\$1.14 at the 5th–95th percentile. Shower energy follows the mixing-fraction physics targeting a fixed 105 °F delivery temperature, so it depends only on the rise from the mains inlet (seasonal inlet temperatures from NREL hot-water modeling [88]) to that target and is independent of heater setpoint; cost is this energy priced at the PA residential rate. Shower environmental impact is defined as water volume (gallons); the realized distribution of flow rate multiplied by duration spans 6.0–45.0 gallons at the 5th–95th percentile, with flow-rate and duration benchmarks from EPA WaterSense Appendix B [46], REU2016 [66], and The Harris Poll [67].

3.3. Ground Truth Calculators

Each ground-truth calculator consumes a scenario with three alternatives and returns four scores per alternative—Energy Cost (\$), Environmental Impact, Comfort, and Practicality (the latter three on 0–10)—plus the raw physical quantities before value-function transformation. Energy cost is always in dollars per 8-hour decision period; Environmental Impact is in lbs CO₂ for HVAC and Appliance decisions and in gallons of water for Shower decisions.

For HVAC and Appliance decisions, environmental impact uses PJM marginal emissions factors:

$$\text{CO}_2 = E_{\text{kWh}} \cdot \varepsilon(t), \quad \varepsilon(t) = \begin{cases} 1.041 \text{ lbs/kWh} & \text{peak (7am–11pm)} \\ 0.976 \text{ lbs/kWh} & \text{off-peak} \end{cases} \quad (2)$$

where E_{kWh} is the decision-period energy use and $\varepsilon(t)$ is the PJM marginal emissions factor at run time t [52]. Marginal rather than average factors are used because shifting a residential load displaces or adds the generator at the margin. All three calculators apply an identical four-tier budget penalty $P(u)$ multiplicatively to the energy-cost value-function output, where $u = C_{\text{monthly}}/B$ is monthly cost utilization relative to the household's stated budget B :

$$P(u) = \begin{cases} 1.0 & u < 0.80 \\ 1 - 2.5(u - 0.80) & 0.80 \leq u < 1.0 \\ 0.5 e^{-3(u-1.0)} & 1.0 \leq u < 1.5 \\ 0 & u \geq 1.5 \end{cases} \quad (3)$$

where $P(u)$ is a unitless multiplicative penalty on the (post value-function) energy-cost score, u is monthly budget utilization, C_{monthly} is estimated monthly energy/water-related cost for the decision type, B is the household-stated monthly budget for utilities, and e is Euler's number. The no-penalty zone ($u < 0.80$) reflects mental budget safety margins [89]; the linear decline approaching the limit follows Heath & Soll's budget self-control model [90]; exponential loss aversion for overruns follows Prelec & Loewenstein [91] and Heutel [92]; and elimination beyond 150% reflects financial infeasibility [93]. Applying the penalty multiplicatively ensures budget-infeasible alternatives cannot dominate MAVT ranking on energy cost alone.

3.3.1. HVAC Calculator

Thermal load decomposes into four ASHRAE-style components. For cooling:

$$Q_{\text{cool}} = \underbrace{\frac{m \cdot A}{R} \Delta T}_{\text{conductive}} + \underbrace{400n + A + 800}_{\text{internal}} + \underbrace{0.15 A \times 20}_{\text{solar}} + \underbrace{1.08 \frac{A h_c \alpha}{60} \Delta T}_{\text{infiltration}} \quad (4)$$

where A is floor area (sq ft); m is a housing-type envelope multiplier (1.7 single-family, 1.6 twin/semi-detached, 1.5 townhouse/rowhouse, 1.2 apartment/condo) [94]; R is the wall insulation R-value; $\Delta T = T_{\text{out}} - T_{\text{in}}$; n is household size; $h_c = 8$ ft and $\alpha = 0.35$ ACH for ceiling height and infiltration rate [94]. The 800 BTU/hr baseline captures always-on standby loads per ACCA Manual J [94]. The heating load uses the same expression with $\Delta T = T_{\text{in}} - T_{\text{out}}$, no solar term, and internal gains subtracted rather than added.

Effective EER is estimated from the unit's rated SEER via the AHRI 210/240 quadratic [95]:

$$\text{EER} = -0.02 \text{SEER}^2 + 1.12 \text{SEER} \quad (5)$$

where SEER is the unit's rated seasonal energy efficiency ratio and EER is the resulting effective energy efficiency ratio used for the decision period. Age- and maintenance-driven efficiency degradation is deliberately *not* applied to the energy path; it is instead modeled as a reliability factor in the practicality score (Eq. 9), so the energy score reflects the setpoint choice rather than the system's condition. Energy consumption over the 8-hour decision window is then:

$$E_{\text{kWh}} = \frac{Q_{\text{load}}}{\text{EER} \times 1000} \cdot h \cdot m_{\text{occ}} \quad (6)$$

where Q_{load} is the thermal load (BTU/hr) for the decision context, $h = 8$ hr is the decision-window duration, and m_{occ} scales for occupancy context: 1.0 for fully occupied, 0.75 for overnight, and $1 - 0.5(h_{\text{away}}/24)$ for daytime unoccupied periods [96], where h_{away} is hours-away per 24-hour day.

Because HVAC alternatives carry no explicit run-time, the marginal emissions factor in Eq. 2 is resolved from the occupancy context by mapping each pattern onto the PJM peak (7am–11pm, 16 h) and off-peak (8 h) windows, with away-hours assumed to fill the daytime peak window first:

$$\varepsilon_{\text{occ}} = \begin{cases} \varepsilon_{\text{off}} & \text{occupied-sleep (night-only runtime)} \\ \frac{16 \varepsilon_{\text{pk}} + 8 \varepsilon_{\text{off}}}{24} & \text{occupied all day} \\ \varepsilon_{\text{pk}} & \text{unoccupied } H \text{ hr, } H \leq 16 \\ \frac{16 \varepsilon_{\text{pk}} + (H - 16) \varepsilon_{\text{off}}}{H} & \text{unoccupied } H \text{ hr, } H > 16 \end{cases} \quad (7)$$

where $\varepsilon_{\text{pk}} = 1.041$ and $\varepsilon_{\text{off}} = 0.976$ lbs/kWh are the PJM peak and off-peak marginal factors and H is hours-away per day [52]. Occupied-sleep returns the off-peak factor because the home’s conditioning runtime falls entirely in the overnight off-peak window.

Comfort uses a tent function anchored to ASHRAE 55 PMV-optimal setpoints [97, 98, 99]:

$$C_{\text{comfort}} = \max(0, 10 - |T_{\text{in}} - T^*|) \quad (8)$$

where C_{comfort} is a 0–10 comfort score, T_{in} is the indoor setpoint, T^* is the context-dependent optimal setpoint used as the peak of the tent function (76 °F in cooling mode, 70 °F in heating mode, consistent with ASHRAE 55-2020 sedentary PMV bands [97]). A small penalty scales with household size beyond three occupants. Practicality starts at 10 and subtracts an asymmetric extremity penalty for setpoints beyond adoption-comfort bounds (in cooling, ≥ 82 °F or ≤ 71 °F; in heating, ≤ 63 °F or ≥ 76 °F), with the steepest per-°F slopes on the least-tolerable directions (1.5/1.0 for the high/low side in cooling; 1.8/0.8 for the low/high side in heating). The result is then multiplied by an outdoor–indoor ΔT feasibility factor (1.0, 0.95, 0.85, 0.70 for $\Delta T < 10, < 20, < 35, \geq 35$ °F), signaling operation near system limits. Finally, age- and maintenance-driven efficiency degradation enters here as a reliability factor (item relocated from the energy path): an older or poorly maintained unit is a less practical choice to rely on, so the score is scaled by $(1 - D)$ where

$$D = \min(0.30, \rho(\text{maint})[1.5 \min(a, 10) + 0.5 \max(a - 10, 0)]) \quad (9)$$

is the fraction of efficiency lost (capped at 30%), a is HVAC age in years, and $\rho(\text{maint})$ is the base annual loss rate (0.005, 0.010, 0.015 per year for good, moderate, and poor upkeep); the first ten years degrade at 1.5ρ and later years at 0.5ρ [100, 101, 102]. The final practicality score is clipped to [1.5, 10].

An “Off” alternative (no active conditioning) is scored with zero decision-period energy, cost, and emissions; its comfort and practicality are instead evaluated against the free-float indoor temperature the building drifts to with the system off — outdoor +5 °F in cooling season and +10 °F in heating season, reflecting solar and internal gains that hold a closed, occupied house above outdoor air temperature [73, 97, 103, 104].

3.3.2. Appliance Calculator

Energy cost per cycle is $C = E_{\text{cyc}} \cdot r(t, \ell)$, where E_{cyc} is kWh/cycle and $r(t, \ell)$ is the location- and time-specific TOU rate resolved to one of six Pennsylvania utilities (PECO, PPL, West Penn, Penelec, MetEd, Duquesne) [53]. where C is per-cycle energy cost (dollars/cycle), E_{cyc} is per-cycle energy use (kWh/cycle), t is the appliance run time (time-of-day), and ℓ is the household location used to resolve the serving utility. The PJM emissions window (7am–11pm) applies to environmental impact regardless of the utility’s billing window, because emissions follow the regional grid. The run-time rate period is selected by mapping location ℓ to a utility $U(\ell)$ and then applying that utility’s TOU peak window otherwise.

$$(h_{\text{pk},s}^U, h_{\text{pk},e}^U) : r(t, \ell) = r_{\text{pk}}^U \text{ if } h(t) \in [h_{\text{pk},s}^U, h_{\text{pk},e}^U) \text{ and } r(t, \ell) = r_{\text{off}}^U \quad (10)$$

where $U(\ell)$ maps location ℓ to a serving utility, $h_{\text{pk},s}^U$ and $h_{\text{pk},e}^U$ are that utility’s TOU peak-window start and end hours, r_{pk}^U and r_{off}^U are the utility’s peak and off-peak energy prices (\$/kWh), and $h(t)$ maps time t to an hour-of-day on a 24-hour clock. In the implementation, the six utilities use distinct peak windows (e.g., 2–6pm for PECO and PPL, 2–9pm for most FirstEnergy utilities). The appliance calculator does not model weekend or holiday peak-hour pricing; all days use weekday TOU rates. Seasonal peak-window shifts are also unmodeled: PPL’s winter (Dec–May) peak window is 4–8pm versus the 2–6pm summer window applied here; the summer window is used year-round because scenarios carry no season or date field.

Comfort is a piecewise decay from 10 at zero delay, with appliance-specific tolerance ceilings of 12 hr (dishwashers), 8 hr (washers), and 6 hr (dryers) [74, 75], plus an additive noise penalty scaled by housing type for runs after 10pm [105, 106] and a household-size penalty. Delays are computed as the minimum circular distance from the baseline time t_0 on a 24-hour clock (avoiding wrap-around artifacts), i.e.,

$$d(t, t_0) = \min((h(t) - h(t_0)) \bmod 24, (h(t_0) - h(t)) \bmod 24), \quad (11)$$

where $d(t, t_0)$ is delay in hours between run time t and baseline time t_0 , $h(\cdot)$ maps a time to hour-of-day, and $\bmod 24$ indicates wrap-around on a 24-hour clock. The base comfort component uses the following piecewise schedule on

delay hours d :

$$C_{\text{base}}(d) = \begin{cases} 10 & d = 0 \\ 10 - \frac{2}{3}d & 0 < d \leq 3 \\ 8 - \frac{1}{2}(d - 3) & 3 < d \leq 7 \\ 6 - \frac{2}{5}(d - 7) & 7 < d \leq 12 \\ 2 & d > 12 \end{cases} \quad (12)$$

where $C_{\text{base}}(d)$ is the pre-penalty comfort score (0–10) as a function of delay d . The late-night noise penalty is applied when running between 10pm and 7am and the appliance exceeds a dBA threshold:

$$N(t, \text{type}, \text{housing}) = \begin{cases} 0 & \text{if } h(t) \in [7, 22) \text{ or } \text{dBA}(\text{type}) \leq \theta \\ 2 \cdot \kappa(\text{housing}) & \text{otherwise} \end{cases} \quad (13)$$

where $N(t, \text{type}, \text{housing})$ is a nonnegative noise-disruption penalty in comfort-score points, $\text{dBA}(\text{type})$ is the appliance noise level for the appliance type, $\theta = 45$ dBA is the dBA threshold, and $\kappa(\text{housing})$ is a unitless housing-type multiplier (larger for shared-wall housing). The household-size penalty scales with delay relative to the appliance-specific maximum acceptable delay $d_{\text{max}}(\text{type})$:

$$S(d, n, \text{type}) = \sigma(n) \cdot \min\left(\frac{d}{d_{\text{max}}(\text{type})}, 1\right), \quad (14)$$

where $S(d, n, \text{type})$ is a nonnegative household-size penalty in comfort-score points, n is occupants, $d_{\text{max}}(\text{type})$ is the maximum acceptable delay (hours) for the appliance type, and $\sigma(n)$ is an occupants-based factor (higher for larger households). Final comfort is then $C_{\text{comfort}} = \text{clip}_{[0,10]}(C_{\text{base}} - N - S)$. where $\text{clip}_{[a,b]}(x) = \min(b, \max(a, x))$ clamps a value to the interval $[a, b]$. Practicality mirrors the delay schedule with additional penalties for late-night timing complexity and household coordination difficulty. The base practicality component is:

$$P_{\text{base}}(d) = \begin{cases} 10 & d = 0 \\ 10 - d & 0 < d \leq 2 \\ 8 - \frac{3}{4}(d - 2) & 2 < d \leq 4 \\ 6.5 - \frac{1}{2}(d - 4) & 4 < d \leq 8 \\ 4.5 - \frac{3}{8}(d - 8) & 8 < d \leq 12 \\ 1.5 & d > 12 \end{cases} \quad (15)$$

where $P_{\text{base}}(d)$ is the pre-penalty practicality score (0–10) as a function of delay d . Timing complexity applies an additive penalty by run-time hour:

$$T(t) = \begin{cases} 2 & 0 \leq h(t) < 6 \\ 1 & 22 \leq h(t) < 24 \\ 0 & \text{otherwise} \end{cases} \quad (16)$$

where $T(t)$ is a nonnegative timing-complexity penalty in practicality-score points and $h(t)$ is hour-of-day. Household coordination difficulty uses the same delay scaling form as in comfort:

$$K(d, n, \text{type}) = \kappa_n(n) \cdot \min\left(\frac{d}{d_{\text{max}}(\text{type})}, 1\right), \quad (17)$$

where $K(d, n, \text{type})$ is a nonnegative household-coordination penalty in practicality-score points and $\kappa_n(n)$ is an occupants-based factor increasing with occupants. Final practicality is $C_{\text{prac}} = \text{clip}_{[1.5,10]}(P_{\text{base}} - T - K)$, with a daytime floor

applied for $h(t) \in [7, 22]$ and $d \leq d_{\max}(\text{type})$. For budget utilization in (3), the appliance calculator converts per-cycle cost to a monthly estimate as:

$$C_{\text{monthly}} = c(\text{type}) C_{\text{cycle}}, \quad c(\text{type}) = \begin{cases} 18 & \text{dishwasher} \\ 25 & \text{washer} \\ 24 & \text{dryer} \end{cases} \quad (18)$$

where C_{cycle} is per-cycle energy cost and $c(\text{type})$ is the appliance-specific cycles-per-month estimate, from 215, 300, and 283 cycles/year for dishwashers, washers, and dryers respectively [47, 107, 108].

3.3.3. Shower Calculator.

Mains inlet temperature is interpolated from outdoor temperature between 45 °F (winter, $T_{\text{out}} \leq 32$ °F) and 65 °F (summer, $T_{\text{out}} \geq 75$ °F) [88, 109]. Only the fraction of flow requiring heating reaches the heater; that mixing fraction is:

$$f_{\text{hot}} = \frac{T_{\text{target}} - T_{\text{inlet}}}{T_{\text{heater}} - T_{\text{inlet}}} \quad (19)$$

where f_{hot} is the fraction of total flow drawn as hot water, $T_{\text{target}} = 105$ °F is the standard comfortable delivery temperature [110], T_{inlet} is mains inlet temperature, and T_{heater} is the water-heater setpoint temperature. Shower energy is then:

$$E_{\text{kWh}} = \frac{\dot{V} \cdot f_{\text{hot}} \cdot 8.33 \cdot (T_{\text{heater}} - T_{\text{inlet}}) \cdot \tau}{3412 \cdot \eta} \quad (20)$$

where E_{kWh} is shower energy use (kWh), $\dot{V}$ is total flow rate (GPM), τ is shower duration (min), $\eta = 0.92$ is the unified energy factor for a 40–55 gal electric tank [111], and 3412 BTU/kWh is the standard conversion; the constant 8.33 lb/gal converts gallons to pounds of water. Environmental impact is water volume: $V = \dot{V} \cdot \tau$ (gallons). Energy cost uses a flat electricity rate p_e :

$$C_{\$} = p_e E_{\text{kWh}}. \quad (21)$$

where $C_{\$}$ is shower energy cost (dollars) and p_e is the electricity price (\$/kWh). Comfort is a piecewise function peaking at the REU2016 average of 7.8 min [66, 67], with additive penalties for temperature adequacy (CDC Legionella thresholds [76]) and household contention. The implementation uses a duration-based piecewise base comfort $C_{\text{base}}(\tau)$ (with a short-duration cubic ramp to reflect severely rushed showers), then subtracts a temperature penalty $\Delta_{\text{temp}}(T_{\text{heater}})$ and a household contention penalty $\Delta_{\text{cont}}(\tau, n)$, and clips to $[0, 10]$. The comfort-optimal duration is the 7.8 min REU2016 mean at moderate outdoor temperatures but is shifted upward in cold weather (to at most ≈ 10.3 min, the observed winter field-mean envelope), reflecting that comfortable showering lengthens as it gets colder; this cold-weather elasticity is an explicit modeling assumption supported only in direction by the literature [66, 67], and warm-weather scoring is unchanged:

$$C_{\text{comfort}}(\tau, T_{\text{heater}}, n) = \text{clip}_{[0, 10]}(C_{\text{base}}(\tau) - \Delta_{\text{temp}}(T_{\text{heater}}) - \Delta_{\text{cont}}(\tau, n)). \quad (22)$$

where τ is shower duration (min), n is occupants, $C_{\text{base}}(\tau)$ is the base comfort curve, $\Delta_{\text{temp}}(T_{\text{heater}})$ is a temperature-adequacy penalty, and $\Delta_{\text{cont}}(\tau, n)$ is a household-contention penalty. Practicality additionally penalizes alternatives that exceed available tank capacity. Unlike comfort, the base practicality curve is not centered on the 7.8 min comfort optimum: it rises with duration up to a peak near 12 min before declining, because practicality here measures *adoptability*—how little behavior change an alternative demands of a household that already showers at a given length—rather than momentary preference. A very short shower scores low on practicality (it requires sustained restraint), while a moderately long one is easy to sustain. Pressure toward shorter showers in the MAVT aggregate therefore comes from the cost and environmental criteria and from the comfort optimum, not from practicality; the capacity constraint below is what makes an over-long shower infeasible for a constrained household. The practicality model uses a duration-based base practicality $P_{\text{base}}(\tau)$ and a capacity constraint computed from hot-water demand. Hot-water per shower is $\tau \dot{V} f_{\text{hot}}$ (gal), total household hot-water need is $n \tau \dot{V} f_{\text{hot}}$, and available draw is approximated as 0.8 of nominal tank size S :

$$H_{\text{need}} = n \tau \dot{V} f_{\text{hot}}, \quad H_{\text{avail}} = 0.8 S. \quad (23)$$

where H_{need} is total hot-water volume required for the household’s showers over the relevant period (gal), H_{avail} is available hot-water draw volume (gal), and S is nominal tank capacity (gal). A fixed penalty is applied when $H_{\text{need}} > H_{\text{avail}}$, yielding a clipped final practicality score. For budget utilization in (3), shower monthly cost is estimated from per-shower cost using an average showers-per-person-per-day rate:

$$C_{\text{monthly}} = C_{\text{shower}} \cdot (n \cdot 0.9 \cdot 30). \quad (24)$$

where C_{shower} is per-shower energy cost, n is occupants, 0.9 is showers/person/day, and 30 is days/month.

3.4. Architecture Implementations

All three architectures call the same underlying model via OpenRouter at temperature 0.3 [112], with model held constant within each benchmark run so that architecture differences rather than model differences drive the comparison. The study evaluates four models spanning a range of capability, reasoning support, and cost: GPT-OSS-20B (reasoning, low effort), Qwen 3.5 9B (non-reasoning), DeepSeek V4 Flash (hybrid reasoning), and Gemini 3.5 Flash (reasoning, minimal effort). Each architecture is described below; pipeline schematics appear in Figure ??.

3.4.1. $\mathcal{A}_D$ (Direct Prompting)

The $\mathcal{A}_D$ architecture issues one API call per alternative (three calls per scenario) using a calibrated system prompt that combines role-prompting [113], structured-output constraints [114], and explicit bias-mitigation language drawn from the NIST framework for identifying and managing AI bias [115]. The model is asked to return JSON with all four criterion scores on a 0–10 scale; no external context is supplied beyond the scenario description.

3.4.2. $\mathcal{A}_E$ (RAG-Enhanced Scoring)

The $\mathcal{A}_E$ architecture retrieves the top $k = 3$ most similar scenarios from a ChromaDB vector store before scoring, using the sentence-transformers/all-MiniLM-L6-v2 embedding model. Each retrieved exemplar is rendered as a full worked example: its homeowner-facing parameters, its withheld engineering parameters (R-value, SEER, GPM, kWh/cycle, tank size, heater setpoint, etc.), and each of its alternatives with the four ground-truth criterion scores plus the MAVT aggregate and rank. This is deliberate—the exemplar’s engineering values are shown so the model can see how those values map to scores—and it does not leak the *target* scenario’s withheld values, which are never placed in the prompt. The model is instructed to use the exemplars as reference but to score the target scenario independently. Like $\mathcal{A}_D$, $\mathcal{A}_E$ uses three API calls per scenario (one per alternative). The retrieval index is built once from the 90 RAG-only scenarios described in Section 2.1 and is never queried with test scenarios [33].

3.4.3. $\mathcal{A}_H$ (LLM-Parameterized Reference Scoring)

The $\mathcal{A}_H$ architecture issues a single API call per scenario for structured parameter extraction. Known homeowner-visible scenario fields (outdoor temperature, square footage, insulation tier, household size, housing type, utility budget, and location) are passed directly to the calculator without LLM involvement, and the three alternatives are taken verbatim from the scenario sheet—the deterministic calculator, not the LLM, parses the numeric content of each alternative string (setpoint temperature, run time, or shower duration). The LLM is asked only to estimate the genuinely withheld engineering parameters that are absent from the homeowner-visible description: for HVAC, the wall R-value, SEER rating, HVAC age, and occupancy context; for Appliance, the appliance type, kWh/cycle, and current baseline time; for Shower, the numeric GPM (estimated from the homeowner-facing flow-rate label), tank size, and water-heater setpoint. The extracted JSON is validated—each numeric parameter must parse as a finite value within its physically admissible range, and the extracted decision type must match the scenario’s known type, or the scenario is recorded as an extraction failure—and then merged with the known fields and passed to the corresponding ground-truth calculator. Because the alternatives and all homeowner facts are supplied deterministically and only the scenario-level engineering parameters are model-estimated, the LLM’s output cannot inject a per-alternative scoring artifact; the calculator does the physics identically to the ground truth. $\mathcal{A}_H$ uses 1 API call per scenario versus 3 for Pure and RAG, exploiting decomposed prompting [116] to separate natural-language understanding from quantitative computation. Because the parameters the LLM estimates are scenario-level and identical across the three alternatives, estimation errors shift all three raw values together and largely cancel in the within-scenario ranking; ranking accuracy is therefore expected to be robust to weak extraction almost by construction, whereas the score-level error (MAE/RMSE) and the per-alternative appliance baseline time are the measures that probe extraction quality.

3.5. RAG Ablation Design

Ablation experiments use a stratified 12-scenario subset (4 per decision type) selected to span the full range of ground-truth scores while keeping computational cost manageable. Six configurations modify one design variable at a time against a $k = 3$ control: retrieval k (1, 5), embedding model (paraphrase-MiniLM-L3-v2 replacing all-MiniLM-L6-v2), exemplar content (hidden engineering parameters included/excluded), and score anchoring (scores and ranks present/absent). Each configuration is evaluated across all four models.

3.6. Evaluation Metrics

Accuracy is evaluated on two axes. Score-level error uses Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) between architecture scores and ground-truth scores on the 0–10 MAVT scale, computed separately for each criterion and aggregated across criteria. Ranking accuracy uses Kendall’s τ and Spearman’s ρ between the architecture’s induced ranking of the three alternatives and the ground-truth ranking, together with Top-1 accuracy (architecture’s best-ranked alternative matches ground truth) and Top-2 accuracy (ground-truth top alternative appears in the architecture’s top two). Efficiency is evaluated using total tokens (input plus output) and total latency, both summed across the 195-scenario test set.

Failure is recorded per scenario via a reserved sentinel value (1928) that marks any malformed output, missing score, out-of-bounds score, or invalid score type, and—for $\mathcal{A}_H$ —any failed parameter extraction (non-JSON output, an invalid or mismatched decision type, or a numeric engineering parameter that does not parse as a finite value in its physical range) or ground-truth calculation error. The accuracy metrics above are therefore *conditioned on success*: a scenario containing any sentinel is excluded from the MAE/RMSE and ranking computations and is instead counted in a separately reported failure rate, so the two are read together (e.g., “ $\tau = x$ over the $1 - f$ fraction that succeeded”). The architectures differ in failure granularity, which we note when comparing rates: $\mathcal{A}_D$ and $\mathcal{A}_E$ issue one call per alternative and mark the whole scenario failed if any one alternative fails, whereas $\mathcal{A}_H$ issues a single extraction call per scenario, so its failure unit is inherently the scenario. Scenario matching between architectures and ground truth uses (question, location) content keys plus decision-type-specific disambiguating parameters, to handle the case where scenario IDs drift across runs.

3.7. End-to-End Workflow

Figure 4 shows the full decision-support workflow common to all three architectures: a natural-language household scenario is converted into MCDA alternatives, criteria, and contextual parameters, scored on the four criteria, and aggregated by the MAVT weighted sum into a ranked recommendation. The three architectures differ only in how the scoring step is performed.

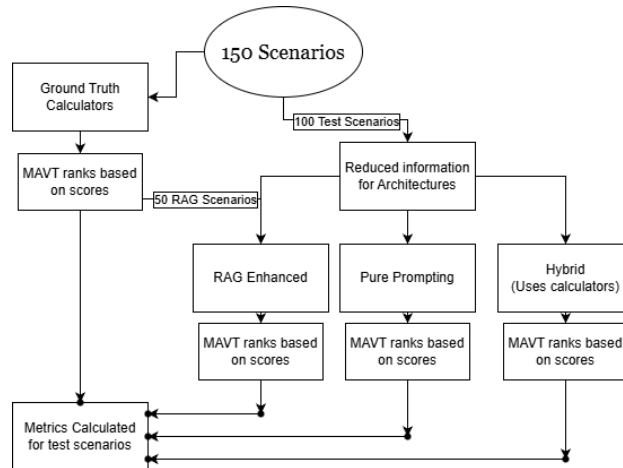


Figure 4: End-to-end MCDA decision-support workflow shared by the three architectures, from natural-language scenario to MAVT-ranked recommendation.

Results are reported as five-run averages: each architecture–model combination is run independently five times with different random seeds, and metrics are computed per run then averaged. Table 11 reports the mean; Figures 10–14 show the per-run distribution.

4. Results

4.1. Overall Ranking Accuracy

Table 11 reports the ranking and error results averaged over five runs per model across 195 test scenarios. Figure 5 shows each model’s trajectory across architectures. $\mathcal{A}_H$ dominates on every metric for all four models: τ ranges from 0.877 (Qwen) to 0.925 (Gemini), Top-1 from 90.3% (Qwen) to 93.3% (Gemini). $\mathcal{A}_E$ places second across all models: τ from 0.282 (Qwen) to 0.378 (Gemini), Top-1 from 49.7% (DeepSeek) to 54.4% (Gemini). $\mathcal{A}_D$ ranks third: τ from -0.002 (Qwen) to 0.169 (Gemini), Top-1 from 31.8% (Qwen) to 40.0% (DeepSeek), near the random baseline of $\tau = 0$ for a three-alternative ranking.

As noted in Section 3.4, the ranking metrics are partly structurally favored in $\mathcal{A}_H$ because extracted parameters are scenario-level; the score-level error (MAE/RMSE) is the stronger evidence of extraction fidelity.

The architecture ordering holds across all four models individually and across all nine weight perturbations in the sensitivity analysis, with τ ranges of 0.901–0.945 for $\mathcal{A}_H$, 0.374–0.463 for RAG, and 0.180–0.323 for Pure. Pairwise differences between architectures are significant under Wilcoxon signed-rank testing ($p < 0.001$ for each adjacent pair).

Per the structural argument in Section 5.1, the score-level error is the stronger evidence of extraction fidelity, shown per model in Figure 6. Across the four models, $\mathcal{A}_H$ MAE ranges from 0.043 (DeepSeek) to 0.068 (Qwen), representing reductions of 63–81% over $\mathcal{A}_D$ (0.208–0.230) and 47–73% over $\mathcal{A}_E$ (0.127–0.159) per model. RMSE follows the same ranking across all four models.

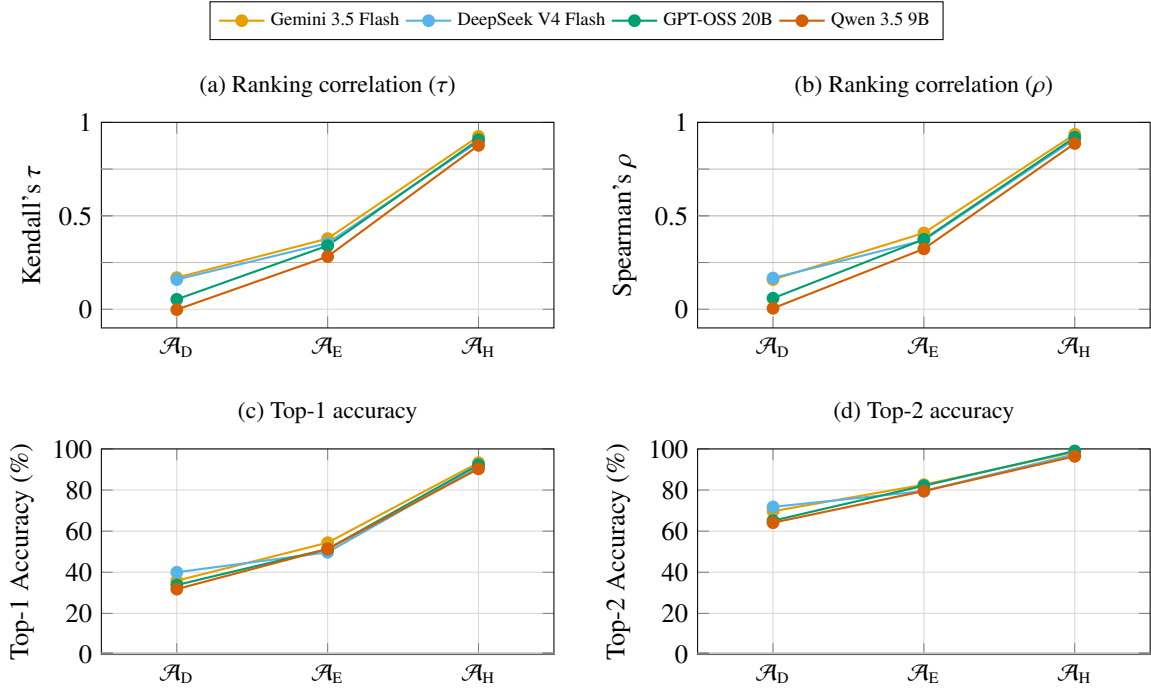


Figure 5: Ranking accuracy by model (5-run mean, 195 scenarios each). Each line is one model; the x-axis is architecture (Pure $\rightarrow$ RAG $\rightarrow$ Hybrid). Panels (a)-(b) and (c)-(d) intentionally use different y-axis scales, since rank correlation and percentage accuracy are not comparable units. Every model’s line rises monotonically left to right, confirming the architecture ordering holds individually for all four models rather than only in a pooled average.

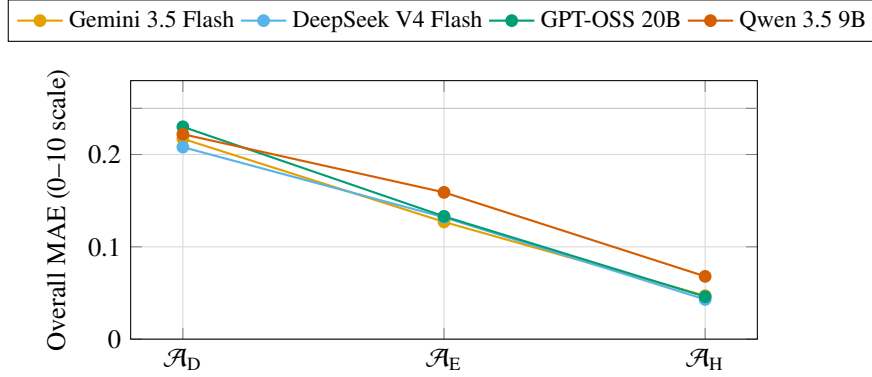


Figure 6: Overall MAE by model (5-run mean, 195 scenarios each). Lower is better; every model’s line descends monotonically from Pure to Hybrid.

4.2. Criterion-Level Error

Table 12 and Figure 7 break MAE down by criterion and model. $\mathcal{A}_H$ achieves near-zero error on Comfort and Practicality across all four models: Comfort MAE ranges from 0.008 (Qwen) to 0.013 (DeepSeek), Practicality from 0.016 (Qwen) to 0.022 (GPT-OSS). This is structural: the calculator handles both criteria deterministically, with the LLM only estimating scenario-level parameters (occupancy, tank size) that shift all three alternatives together.

Error for all three architectures concentrates on Energy Cost and Environmental Impact. $\mathcal{A}_D$ Environmental MAE spans 0.245–0.285 across models and Energy Cost 0.219–0.249. $\mathcal{A}_E$ reduces these to 0.131–0.169 (Environmental) and 0.137–0.171 (Energy Cost). $\mathcal{A}_H$ cuts further to 0.076–0.141 (Environmental) and 0.063–0.105 (Energy Cost); Qwen carries the largest residual on both, indicating its smaller parameter count produces less precise domain estimates. The Environmental–Energy Cost gap persists across all models: $\mathcal{A}_D$ Environmental MAE exceeds Energy Cost for all four models, indicating LLMs struggle more with emissions estimation than with dollar costs.

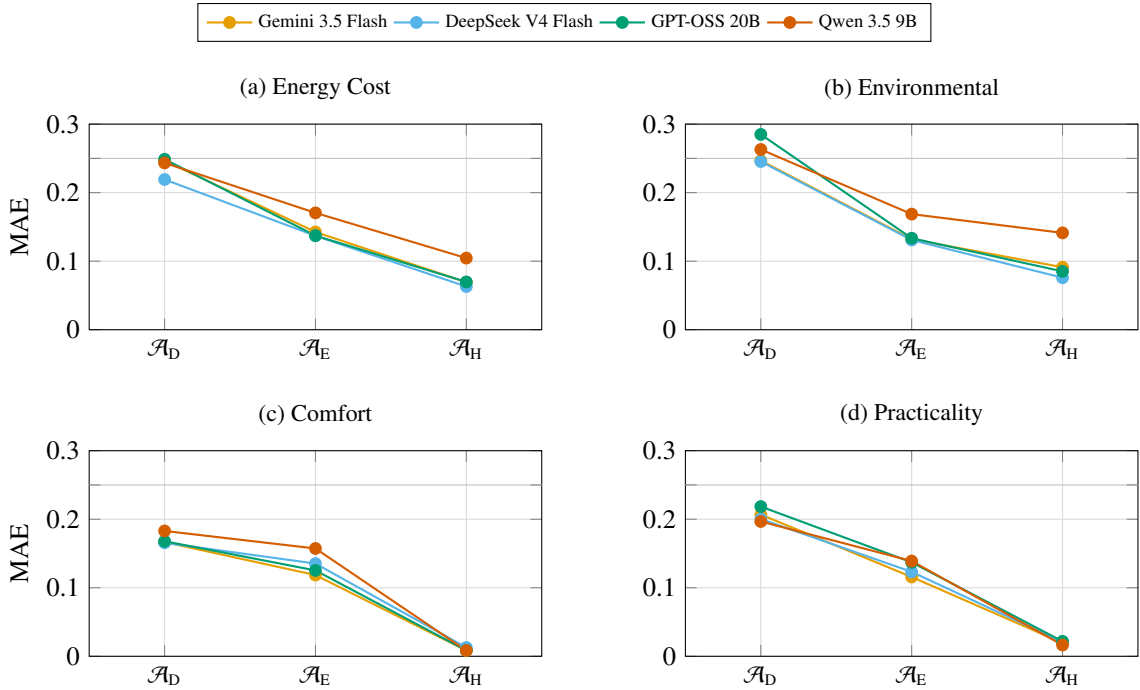


Figure 7: MAE by criterion and model (5-run mean, 195 scenarios each). Each line is one model; the x-axis is architecture (Pure → RAG → Hybrid).

4.3. Performance by Decision Type

Table 13 and Figure 8 report τ by decision type, with each line in Figure 8 representing one model. $\mathcal{A}_H$ excels across all three decision types for every model. On HVAC, individual model τ values range from 0.876 (Qwen) to 0.981 (Gemini). On Appliance, three models reach $\tau = 0.959$ and Qwen reaches $\tau = 0.897$. Shower is the weakest type due to GPM estimation limits, with model τ from 0.789 (DeepSeek) to 0.856 (Qwen), still far above the other architectures.

$\mathcal{A}_E$ shows wide model-level variance on HVAC: Gemini ($\tau = -0.076$) and DeepSeek ($\tau = 0.048$) are near-random, while GPT-OSS ($\tau = 0.267$) and Qwen ($\tau = 0.191$) achieve modest positive correlation. On Appliance, Gemini ($\tau = 0.456$) and DeepSeek ($\tau = 0.364$) lead; GPT-OSS ($\tau = 0.067$) and Qwen ($\tau = 0.241$) lag. Shower is the strongest type for $\mathcal{A}_E$ across all models: Gemini ($\tau = 0.822$), GPT-OSS ($\tau = 0.722$), DeepSeek ($\tau = 0.700$), and Qwen ($\tau = 0.433$).

$\mathcal{A}_D$ is near-random or below on HVAC for all four models: Gemini -0.048 , DeepSeek -0.124 , GPT-OSS 0.238 , Qwen 0.057 . All four models score negative on Appliance (range -0.190 to -0.026). Shower diverges: Gemini (0.633) and DeepSeek (0.722) show meaningful positive correlation, while GPT-OSS (0.100) and Qwen (0.067) remain near-random. The shower advantage for $\mathcal{A}_D$ concentrates in Gemini and DeepSeek, likely because shower scoring depends on duration and flow rate, which are easier to read from scenario text than HVAC load calculations or appliance TOU scheduling.

Appliance results for $\mathcal{A}_H$ track extraction quality since the baseline time parameter varies across alternatives, making it the strongest test of extraction fidelity.

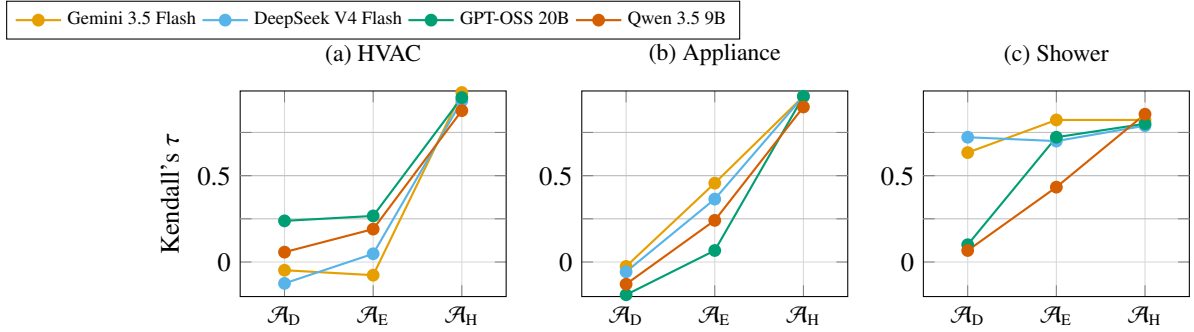


Figure 8: Kendall’s τ by decision type and model (5-run mean, 195 scenarios each). Each line is one model; the x-axis is architecture (Pure $\rightarrow$ RAG $\rightarrow$ Hybrid). The dotted line marks $\tau = 0$ (random ranking).

4.4. RAG Ablation

Table 15 summarizes a RAG ablation study conducted on a 12-scenario subset with six configurations across five models. The nearest-neighbor offline baseline assigns retrieved scenario scores directly without LLM rescoring and yields $\tau = 0.000$, confirming that naive exemplar copying provides no ranking signal.

Retrieval $k = 1$ ($\tau = 0.341$) matches the $k = 3$ control ($\tau = 0.339$), and the exemplars_no_hidden_params configuration ($\tau = 0.337$) performs comparably. The alternate embedding model (paraphrase-MiniLM-L3-v2) provides a modest improvement ($\tau = 0.342$) and a larger gain on Appliance ($\tau = 0.733$ versus 0.690 for the control). The descriptions_no_scores_ranks configuration, which omits all quantitative scores from the retrieved exemplars, performs worst ($\tau = 0.182$), confirming that exemplar scores provide critical anchoring information.

4.5. Efficiency and Cost

Table 14 reports API usage and costs. $\mathcal{A}_H$ is both the cheapest and the most accurate architecture, requiring one API call per scenario versus three for $\mathcal{A}_D$ and $\mathcal{A}_E$. Cost per 195-scenario run ranges from \$0.01 (GPT-OSS, Qwen) to \$0.28 (Gemini). $\mathcal{A}_D$ costs 2–6 $\times$ more per run (\$0.02–\$0.80). $\mathcal{A}_E$ is the most expensive, driven by the token overhead of three retrieved exemplars per call (\$0.03–\$1.28).

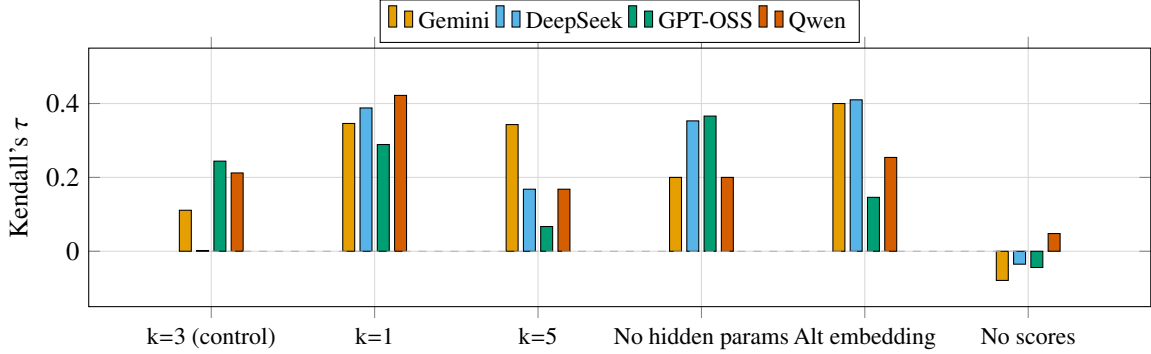


Figure 9: RAG ablation results: Kendall’s τ by configuration and model on the 12-scenario subset.

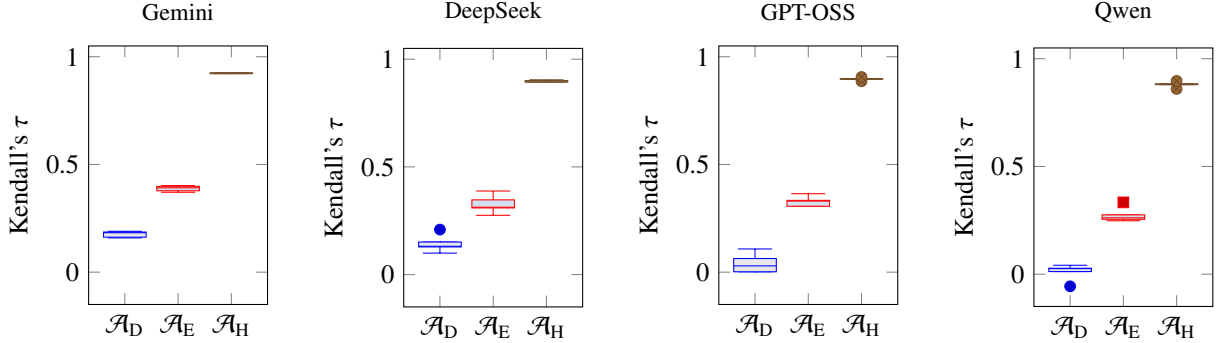


Figure 10: Distribution of Kendall’s τ across five independent runs per model. Box plots show median, quartiles, and whiskers.

Failure rates are near-zero across all architectures and models. The single exception is GPT-OSS $\mathcal{A}_H$, which shows a 12.8% extraction failure rate on parameter parsing. Multi-run averaging recovers these scenarios, yielding 0% scenario-level failure after matching.

4.6. Run-to-Run Variance

The results above are five-run averages. To assess trial-to-trial stability, we computed every metric independently for each of the five runs per architecture per model. Figures 10–14 show the distributions.

RMSE, which penalizes large errors more heavily than MAE, shows the same architecture separation. $\mathcal{A}_H$ RMSE is < 0.11 for every model and every run, while $\mathcal{A}_D$ never drops below 0.29. The Qwen $\mathcal{A}_H$ box shows a wider interquartile range (RMSE $\in [0.157, 0.166]$), reflecting the GPM estimation imprecision that also limits its τ . Across all five metrics and all four models, the architecture ordering $\mathcal{A}_H > \mathcal{A}_E > \mathcal{A}_D$ holds in every single run—there is no run in which any architecture’s median metric overtakes another’s.

5. Discussion

5.1. Limitations

The benchmark is run over four models spanning a range of capability and open/closed weights—GPT-OSS 20B (reasoning, low effort), Qwen 3.5 9B (non-reasoning), DeepSeek V4 Flash (hybrid reasoning), and Gemini 3.5 Flash (reasoning, minimal effort)—all called through OpenRouter at temperature $T = 0.3$ [112], with the model held constant within each benchmark run so architecture differences rather than model differences drive each comparison. The architecture ordering ($\mathcal{A}_H > \text{RAG} > \text{Pure}$) holds for all four model families, confirming that the $\mathcal{A}_H$ advantage is not model-specific. The per-decision-type pattern reported in Section 4.3 also persists across models: $\mathcal{A}_H$ achieves

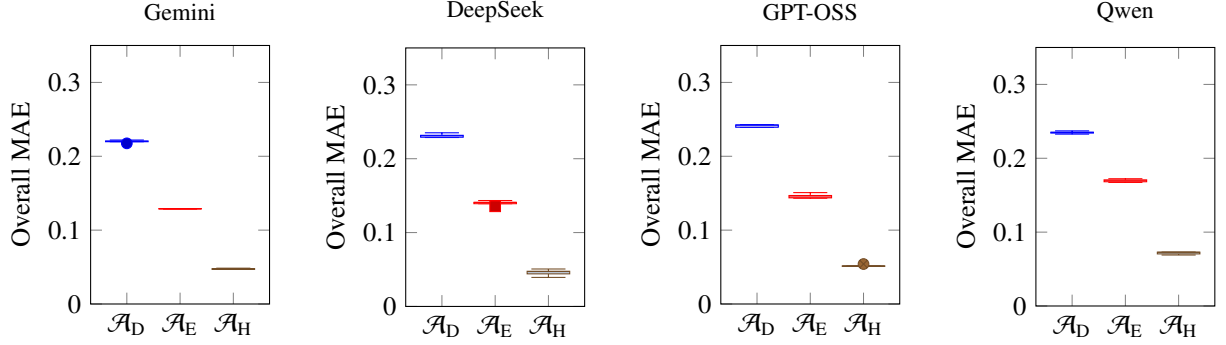


Figure 11: Distribution of Overall MAE across five independent runs per model.

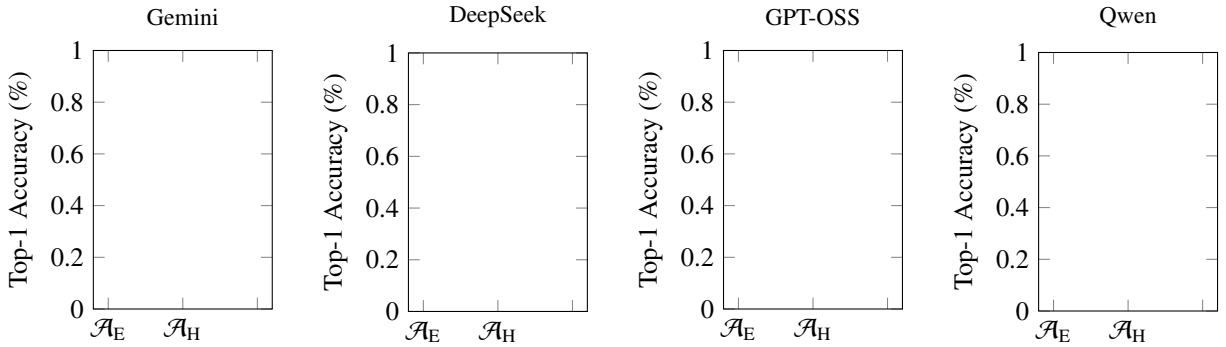


Figure 12: Distribution of Top-1 Accuracy across five independent runs per model.

$\tau > 0.9$ on HVAC and Appliance for every model, while Shower τ ranges from 0.800 to 0.850 due to GPM estimation precision limits. $\mathcal{A}_D$ remains near-or-below-random on HVAC and Appliance for all four models.

We completed five runs per model. Standard deviations across runs are small relative to the between-architecture gaps, confirming statistical significance of the architecture ordering.

The RAG ablation study (Section 4.4) reinforces several findings. The nearest-neighbor baseline performs at random ($\tau = 0.000$), confirming that unmodified exemplar copying provides no ranking signal. The $k = 1$ retrieval matches the $k = 3$ control, suggesting that a single well-matched exemplar can be as effective as multiple distractors. Excluding scores and ranks from exemplars degrades performance sharply, while excluding hidden engineering parameters has less effect, indicating that the scores themselves carry the anchoring information. These ablation results are limited by the small sample (12 scenarios) and should be validated on the full 195-scenario set.

The most critical validity threat is that the same researcher designed both the ground truth calculators and the $\mathcal{A}_H$ architecture. $\mathcal{A}_H$'s advantage may partly reflect design alignment—the calculator the $\mathcal{A}_H$ calls is the same one that produces the ground truth—rather than a generalizable architectural superiority. This threat cannot be eliminated within the current study design and must be weighed when interpreting the magnitude of $\mathcal{A}_H$'s performance gap over $\mathcal{A}_E$ and $\mathcal{A}_D$.

A related and more subtle point concerns *what* the $\mathcal{A}_H$ ranking metrics actually test. Every engineering parameter the $\mathcal{A}_H$ LLM estimates (HVAC: R-value, SEER, HVAC age, occupancy context; Appliance: kWh/cycle; Shower: GPM, tank size, heater setpoint) is a *scenario-level* quantity—identical across the three alternatives being ranked within a scenario. Because the calculator applies the same estimated parameter to all three alternatives, an estimation error shifts all three raw values together and largely cancels in the *within-scenario ranking*; it survives only through nonlinearities (value-function clipping, the budget-penalty thresholds, the shower tank-capacity threshold) and through the single per-alternative quantity the LLM influences, the appliance baseline time. Consequently, $\mathcal{A}_H$'s ranking accuracy (τ , Top-1/Top-2) is expected to be robust to weak extraction almost by construction, whereas the score-level error (MAE/RMSE) and the Appliance ranking are the measures that probe extraction quality. We there-

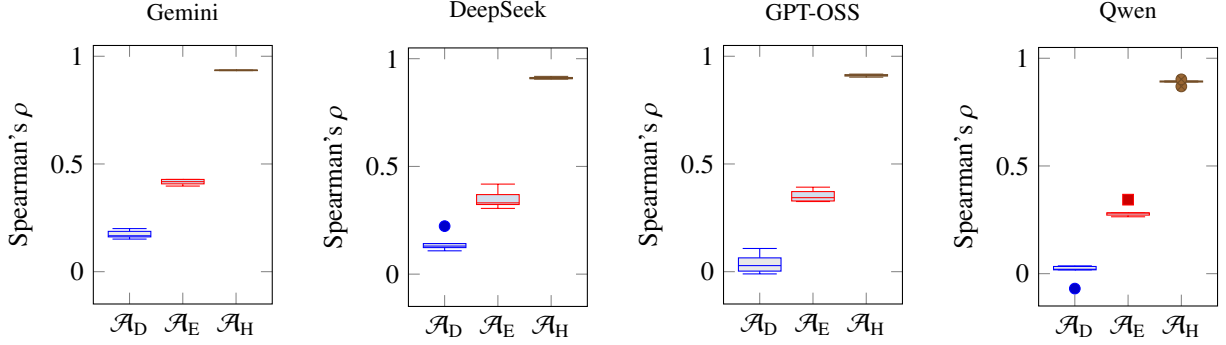


Figure 13: Distribution of Spearman's ρ across five independent runs per model.

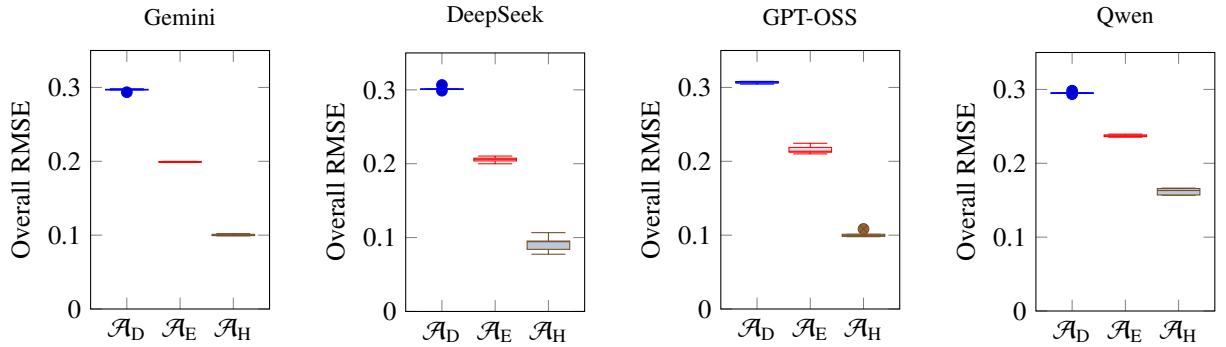


Figure 14: Distribution of Overall RMSE across five independent runs per model.

fore interpret a model-invariant $\mathcal{A}_H$ ranking advantage as partly structural rather than as evidence that weak models extract accurately, and we report the score-level error and the per-decision-type ranking separately so the two effects are not conflated.

Results are specific to Pennsylvania's grid and utility rate structure. The PJM marginal emissions factors (peak: 1.041 lbs CO₂/kWh; off-peak: 0.976 lbs CO₂/kWh) and Pennsylvania time-of-use rate tiers used in the ground truth calculators do not generalize to high-renewables grids, where lower and more variable emissions intensities would compress environmental score distributions and potentially change which architecture performs best.

The environmental criterion for shower decisions uses water consumption in gallons as a proxy rather than CO₂ emissions. This makes the environmental dimension semantically heterogeneous across decision types: HVAC and Appliance environmental scores reflect grid emissions, while Shower environmental scores reflect water volume. Cross-decision-type comparisons of environmental MAE should be interpreted with this distinction in mind; the proxy does not affect within-type ranking validity.

5.1.1. Calculator-Side Imputations

Five engineering simplifications in the ground-truth calculators shape the reference scores. Each affects all architectures equally (since they share the same calculator) and does not alter the within-type ranking comparison, but each does limit the absolute fidelity of the ground truth itself.

Mains inlet temperature.. The shower calculator linearly interpolates mains water temperature from outdoor temperature: $T_{\text{inlet}} = 45^\circ\text{F}$ when $T_{\text{out}} \leq 32^\circ\text{F}$ and $T_{\text{inlet}} = 65^\circ\text{F}$ when $T_{\text{out}} \geq 75^\circ\text{F}$, with linear interpolation between. A household in Philadelphia (outdoor temp 45°F in March) gets $T_{\text{inlet}} = 50^\circ\text{F}$. Actual ground-water temperature from the NREL hot-water model at that outdoor temp is closer to 48°F . The 2°F gap shifts the mixing fraction f_{hot} by roughly 0.015, which changes shower energy cost by about \$0.003 per shower. Across 195 scenarios, the error is small

but systematic: the calculator overestimates inlet temperature in shoulder seasons (April/May, October/November), compressing the energy-cost distribution and nudging environmental scores downward.

Monthly cycle rates.. The appliance budget penalty converts per-cycle cost to monthly cost using fixed rates: 18 cycles/month for dishwashers, 25 for washers, 24 for dryers [47, 107, 108]. A household that runs the dishwasher twice daily (60 cycles/month) faces a monthly cost three times higher than the calculator assumes. The budget penalty $P(u)$ activates when utilization $u = C_{\text{monthly}}/B$ exceeds 0.80, so a household with high actual usage but a low assumed rate may receive a penalty score of 1.0 (no penalty) when the true penalty should be 0.5 or lower. The per-cycle cost itself is computed correctly from the kWh/cycle and TOU rate; the imputation enters only at the monthly extrapolation step.

Hot-water tank draw fraction.. The shower practicality model assumes 80% of nominal tank size S is available for draw: $H_{\text{avail}} = 0.8 S$. A 40-gallon tank yields 32 gallons available. Actual draw fraction depends on tank recovery rate, inlet temperature, and heater power; a high-recovery 50-gallon tank may deliver 90%+ usable volume, while a low-recovery 30-gallon unit may deliver closer to 65%. For a four-person household showering consecutively (needing $4 \times 7.8 \times 2.5 \times 0.7 = 54.6$ gallons), the 0.8 factor produces $H_{\text{need}} > H_{\text{avail}}$ and triggers the capacity penalty. A different factor (say 0.7 or 0.9) would shift the penalty threshold and could flip the ranking between a 7-minute and a 9-minute shower alternative.

Band quantization.. Insulation tiers (Poor/Medium/Good) map to fixed R-values (R-11/R-13/R-19), and appliance age bands (1–15 yr) map to kWh/cycle ranges. A house with R-15 walls receives R-13 (Medium) in both the ground truth and the AH extraction target. The 2 R-value difference changes HVAC load by roughly 4%, which shifts energy cost by about \$0.13 per 8-hour period. Banding compresses a continuous parameter into a discrete set, introducing quantization error that propagates through the value functions. The error is symmetric: the ground truth scores the R-13 value, and the LLM extracts "Medium" which maps back to R-13. Both pipelines see the same quantized input, so the architecture comparison remains valid, but the absolute scores diverge from what a fully continuous parameterization would produce.

Occupancy context.. The HVAC calculator maps occupancy context (fully occupied, overnight, daytime unoccupied) to a run-time fraction: 1.0, 0.75, or $1 - 0.5(h_{\text{away}}/24)$. A household with one occupant away from 9am–5pm (8 hours) gets 0.833. A household with two occupants on different schedules gets the same 0.833 if the aggregate away-hours match. The mapping discards intra-day variation: a household that cycles the HVAC off for 2 hours at noon and on for 2 hours at midnight receives the same fraction as one that holds steady all day. The resulting energy use differs by 5–12% depending on outdoor temperature profile, which shifts energy cost by \$0.05–\$0.25 per period.

Implications for interpretation.. These five imputations introduce systematic error in the ground truth that affects all architectures equally. The architecture ranking ($\mathcal{A}_H > \mathcal{A}_E > \mathcal{A}_D$) is robust to them because each architecture sees the same calculator output. The absolute MAE and RMSE values, however, reflect the combined error of the LLM extraction and the calculator imputation. A perfect LLM extractor would still produce non-zero MAE against the ground truth if the ground truth itself deviates from measured household data. Future work that validates the calculators against instrumented households would quantify this floor and establish the minimum achievable MAE for each architecture.

The prompts have variations between architectures. $\mathcal{A}_D$ includes reference consumption ranges to anchor the LLM’s numeric estimates. $\mathcal{A}_E$ replaces those anchor values with three scored exemplars. $\mathcal{A}_H$ has an entirely different structure: it asks the LLM to estimate engineering parameters rather than to produce scores directly. These differences are inherent to each architecture’s design and may contribute to the observed performance gaps.

A more fundamental constraint concerns the implementation cost of the $\mathcal{A}_H$ architecture itself. The three ground-truth calculators developed for this study—covering HVAC load estimation, appliance TOU scheduling, and shower thermodynamics—each required extensive domain research, iterative calibration, and verification against published standards (ASHRAE, ENERGY STAR, EPA eGRID). In practice, this engineering effort constituted the majority of the total project time. This cost does not diminish $\mathcal{A}_H$ ’s accuracy advantage, but it does bound its practical applicability: $\mathcal{A}_H$ is the appropriate architecture when a vetted, domain-specific calculator already exists or is worth building for the decision at hand. For decision domains where no such calculator exists and the cost of developing one is

prohibitive, $\mathcal{A}_E$ or sufficiently capable $\mathcal{A}_D$ may remain the only accessible options, even at a known accuracy cost. This tradeoff is the primary practical implication of RQ2 and should be weighed carefully before adopting the $\mathcal{A}_H$ approach in new domains.

5.2. Future Work

This study evaluates scoring accuracy only; whether households would actually adopt an LLM-MCDA tool in practice—and the behavioral and workflow factors that govern adoption—is outside its scope and left to future work.

The multi-model benchmark across GPT-OSS 20B, Qwen 3.5 9B, DeepSeek V4 Flash, and Gemini 3.5 Flash confirms that the $\mathcal{A}_H$ advantage is robust across all four models. The architecture ordering ($\mathcal{A}_H > \text{RAG} > \text{Pure}$) holds in every case.

The $\mathcal{A}_H$ per-decision-type τ values are 0.936 (HVAC), 0.944 (Appliance), and 0.817 (Shower). Appliance τ exceeds HVAC by 0.008. The Shower τ is lower due to GPM estimation limits. Scenario-type-specific extraction prompts for Appliance and Shower decisions should be developed, as the current unified extraction prompt was calibrated primarily against HVAC scenarios.

Replacing the water-consumption proxy in the Shower environmental criterion with a lifecycle CO_2 factor—converting gallons of hot water to grid emissions using a standardized heating efficiency—would make the environmental criterion semantically consistent across all three decision types and enable valid cross-type environmental MAE comparisons.

6. Conclusion

This study benchmarked three strategies for integrating an LLM into an MCDA workflow for household energy decisions—direct $\mathcal{A}_D$, $\mathcal{A}_E$ prompting, and $\mathcal{A}_H$ that uses the LLM only to extract withheld engineering parameters and a deterministic physics calculator to score—against a physics-based MAVT ground truth across HVAC, Appliance, and Shower decisions. $\mathcal{A}_H$ achieved Kendall’s τ of 0.902 and MAE of 0.051, compared to 0.339 and 0.138 for $\mathcal{A}_E$ and 0.095 and 0.219 for $\mathcal{A}_D$. The architecture ordering held across all four models.

Grounding the LLM’s numeric reasoning in a deterministic calculator improves agreement with the physics ground truth, and the advantage holds as model capability decreases. A physics backbone compensates for weaker models. We separate this claim into its ranking and score-level components. The ranking robustness of the $\mathcal{A}_H$ design is partly structural (extracted parameters are scenario-level and cancel within a ranking), so the score-level error and the appliance-timing result are the load-bearing measures of extraction fidelity. The two principal threats—one researcher designed both the calculator and the $\mathcal{A}_H$ extractor, and the single-grid (Pennsylvania/PJM) specificity of the ground truth—are stated in Section 5.1 and bound the generalization.

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Arch.	Role	Task	Output	Calibration
$\mathcal{A}_D$	Expert household decision analyst	Score 3 alternatives on 4 criteria	JSON {4 scores, 0–1}	Per-criterion rubrics (good/moderate/poor) + bias
$\mathcal{A}_E$	Expert household decision analyst	Score 3 alternatives using exemplars	JSON {4 scores, 0–1}	$k = 3$ retrieved exemplars with ground-truth scores
$\mathcal{A}_H$	Household-systems engineer	Extract engineering parameters	JSON {parameter values}	Mandatory estimation fields + calculator routing

Table 8: Summary of the three architecture prompt designs. Full prompt strings are available in the Python source files for reproducibility.

Criterion	Weight (%)	Value Function	α	Direction
Environmental Impact	35	Linear	—	Lower raw = better
Energy Cost	30	Linear	—	Lower raw = better
Comfort	20	Logarithmic	1.5	Higher raw = better
Practicality	15	Logarithmic	1.2	Higher raw = better

Table 9: MAVT criterion specification. Weights sum to 100% and are held constant across all architectures and the ground truth calculator.

$$v_i(x) = \frac{x - x_{\min}}{x_{\max} - x_{\min}} \times 10 \quad (25)$$

$$v_i(x) = \frac{\ln\left(1 + \alpha \cdot \frac{x - x_{\min}}{x_{\max} - x_{\min}}\right)}{\ln(1 + \alpha)} \times 10 \quad (26)$$

Decision Type	Test Set (n)	RAG Set (n)	Total (n)	Decision Type	GT Calculator Receives	Withheld / Generalized for AI
				HVAC	R-value, SEER, HVAC age, occupancy context, sq. ft., insulation	R-value, SEER, HVAC age, occupancy context (insulation shown as good/avg/poor)
HVAC	70	35	105	Appliance	kWh/cycle, TOU rates, baseline time, appliance type	kWh/cycle, TOU rates (type and baseline time inferred from question text)
Appliance	65	35	100			
Shower	60	20	80	Shower	GPM, tank size, water heater setpoint	GPM (from flow-rate label), tank size, water heater setpoint
Total	195	90	285			

Table 10: Left: scenario distribution. Right: parameters available to the ground-truth calculator but withheld or generalized for AI input.

Metric	$\mathcal{A}_D$	$\mathcal{A}_E$	$\mathcal{A}_H$
Ranking Accuracy			
Kendall’s τ	0.095	0.339	0.902
Spearman’s ρ	0.097	0.369	0.912
Top-1 Accuracy (%)	35.4	51.7	91.7
Top-2 Accuracy (%)	67.7	80.9	97.8
Score Error			
Overall MAE	0.219	0.138	0.051
Overall RMSE	0.283	0.201	0.105

Table 11: Overall performance comparison (pooled across 4 models, 5 runs each, 195 scenarios).

Criterion	$\mathcal{A}_D$	$\mathcal{A}_E$	$\mathcal{A}_H$
Energy Cost	0.240	0.147	0.077
Environmental Impact	0.260	0.141	0.098
Comfort	0.171	0.134	0.010
Practicality	0.206	0.129	0.020
Overall	0.219	0.138	0.051

Environmental MAE for Shower reflects water-consumption proxy (gallons); see Section 3.3.

Table 12: MAE per criterion by architecture (pooled across 4 models, 5 runs each).

Decision Type	$\mathcal{A}_D$			$\mathcal{A}_E$			$\mathcal{A}_H$		
	τ	Top-1 (%)	Top-2 (%)	τ	Top-1 (%)	Top-2 (%)	τ	Top-1 (%)	Top-2 (%)
HVAC	0.031	32.5	68.9	0.107	33.6	70.4	0.936	92.5	98.9
Appliance	-0.100	25.8	55.4	0.282	53.8	80.8	0.944	96.9	99.2
Shower	0.381	49.2	79.6	0.669	70.4	93.3	0.817	85.4	95.0
Overall	0.095	35.4	67.7	0.339	51.7	80.9	0.902	91.7	97.8

Table 13: Ranking accuracy by decision type (pooled across 4 models, 5 runs each, 195 scenarios).

Architecture	Gemini	DeepSeek	GPT-OSS	Qwen
$\mathcal{A}_D$	\$0.80	\$0.10	\$0.02	\$0.02
$\mathcal{A}_E$	\$1.28	\$0.09	\$0.03	\$0.03
$\mathcal{A}_H$	\$0.28	\$0.01	\$0.01	\$0.01

API Calls per run: Pure=585, RAG=585, $\mathcal{A}_H$ =195 (195 scenarios $\times$ 3 or 1 calls/scenario).

Failure rates: Gemini 3.5 Flash 0%, DeepSeek V4 Flash 0%, GPT-OSS 20B 0% Pure/RAG, 12.8% extraction failures on $\mathcal{A}_H$ (0% scenario failure after multi-run matching), Qwen 3.5 9B 0%.

Table 14: Cost per run (USD), API calls, and failure rates. Costs estimated from OpenRouter pricing.

Configuration	Overall τ	Key Finding
Control ($k=3$, all-MiniLM-L6-v2)	0.339	Baseline RAG configuration
Nearest-neighbor (offline)	0.000	Random-level performance
retrieval_k=1	0.341	Matches $k = 3$ control
retrieval_k=5	0.186	Weaker than $k = 3$ or $k = 1$
Alt. embedding (paraphrase-MiniLM-L3-v2)	0.342	Improves Appliance ($\tau = 0.733$)
exemplars_no_hidden_params	0.337	Comparable to control
descriptions_no_scores_ranks	0.182	Worst configuration

Evaluated on a 12-scenario subset with 5 models. Nearest-neighbor assigns retrieved scenario scores directly without LLM rescoreing.

Table 15: RAG ablation: Kendall’s τ by configuration.